固体表面化学
Chemistry on Solid Surfaces

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参考书

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Chapter 1 Surfaces-An Introduction

1.1 Historical perspectives

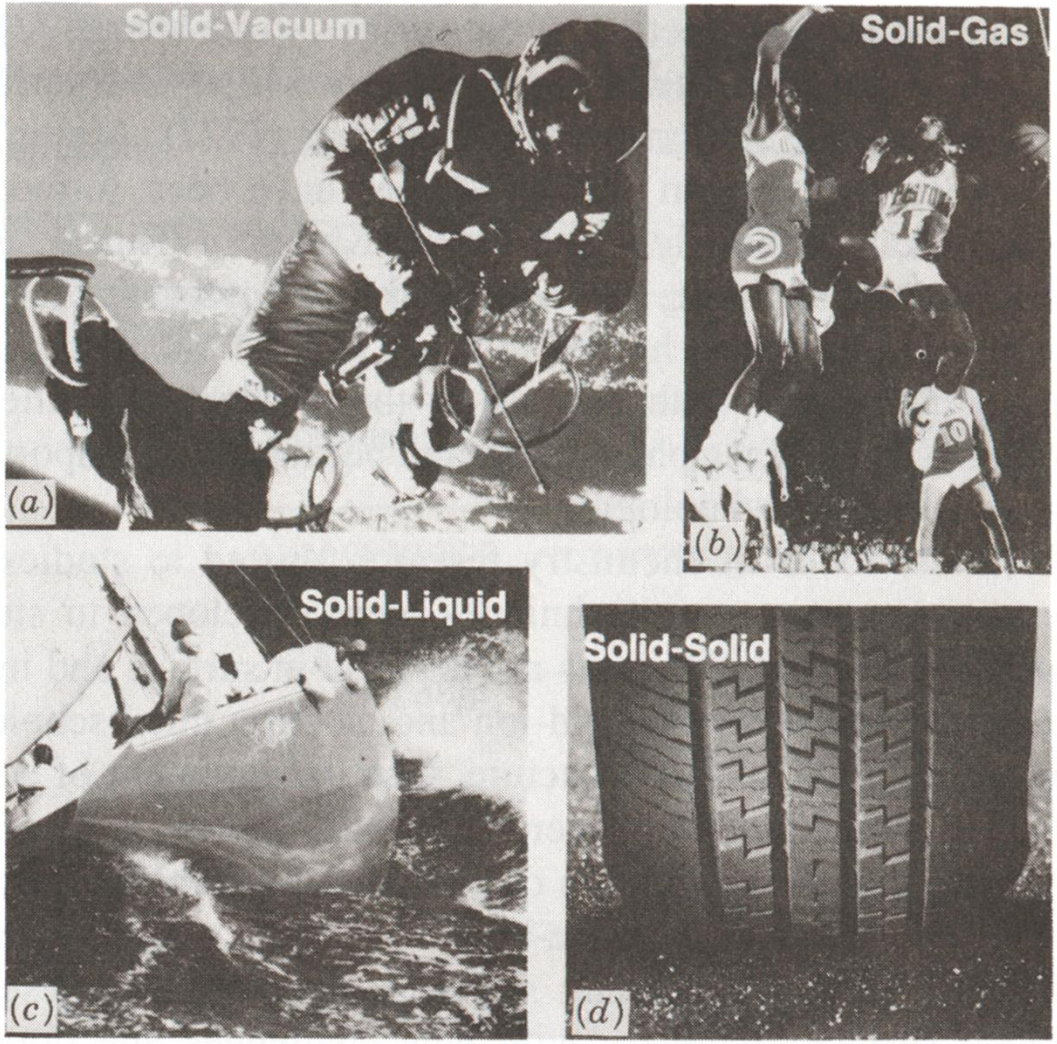
1.2 Concepts related to solid surfaces

1. Surface density
2. Dispersion
3. Thin films
4. Internal surface
5. Adsorption

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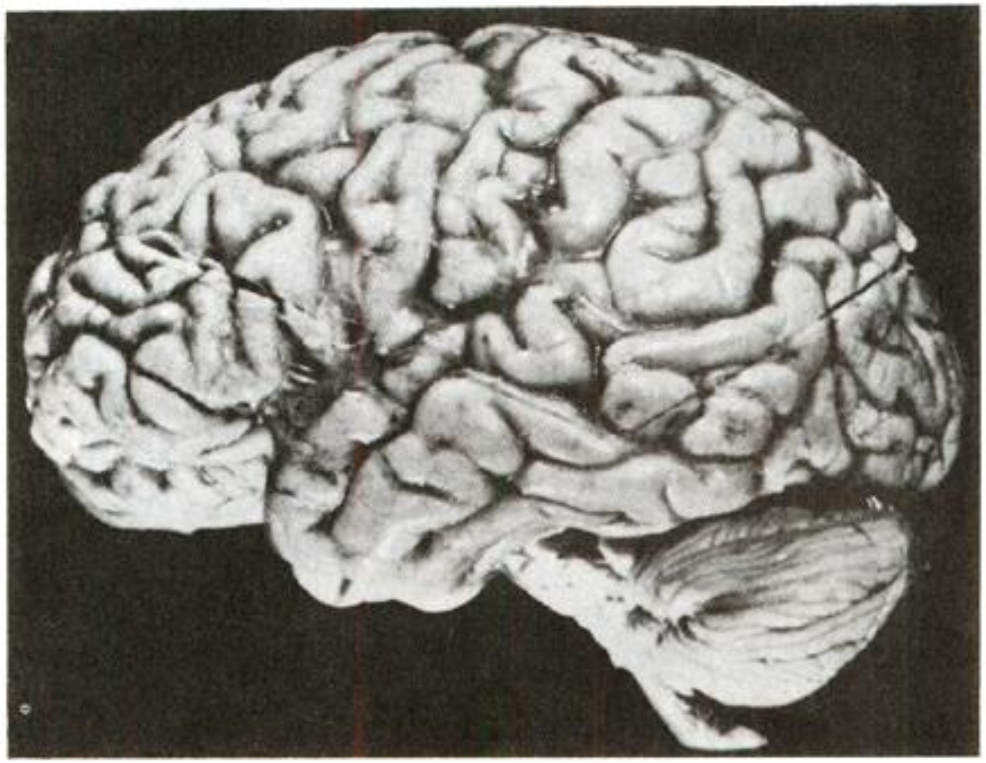
1.4 Methods for surface characterizations

Surface and Interface in nature or our daily life



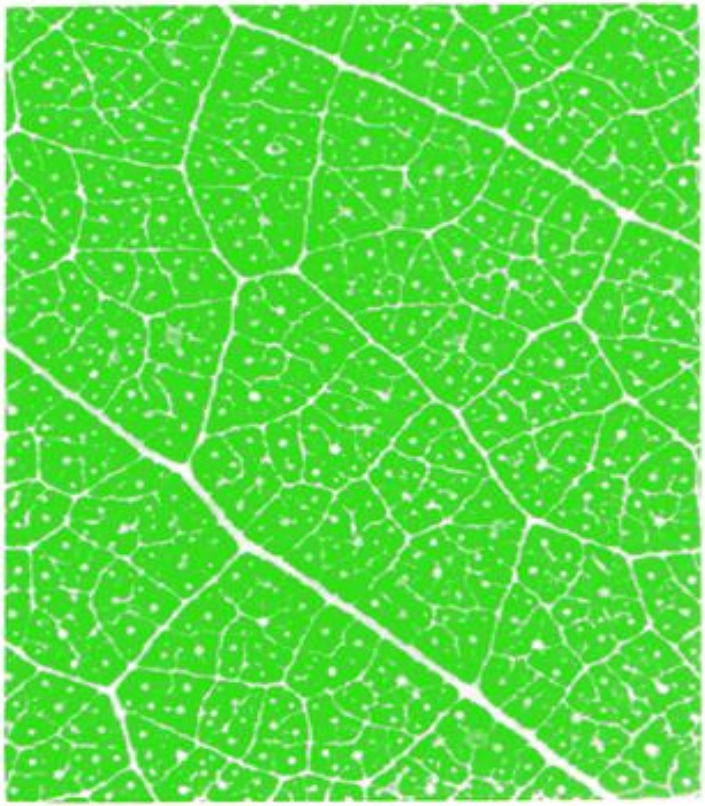
Interfaces are ever-present in our lives. (**a**) An astronaut representing the solid–vacuum interface; (**b**) a jumping basketball player representing the solid–gas interface; (**c**) a sailboat representing the solid–liquid interface; and (**d**) a tire representing the solid–solid interface.

Surface and Interface in nature or our daily life



Human brain The intricate folds of the human brain expose the large interface area of this remarkable organ. The brain may be viewed as a device with enormous solid–liquid interface area.

Surface and Interface in nature or our daily life



Surface for phtosynthesis

The green leaf. Photosynthesis involves the absorbtion of sunlight and the reactions of water and carbon dioxide to produce organic molecules and oxygen. High-surface-area systems, such as the green leaf, are most efficient to carry out photosynthesis.

1.1 Historical Perspective

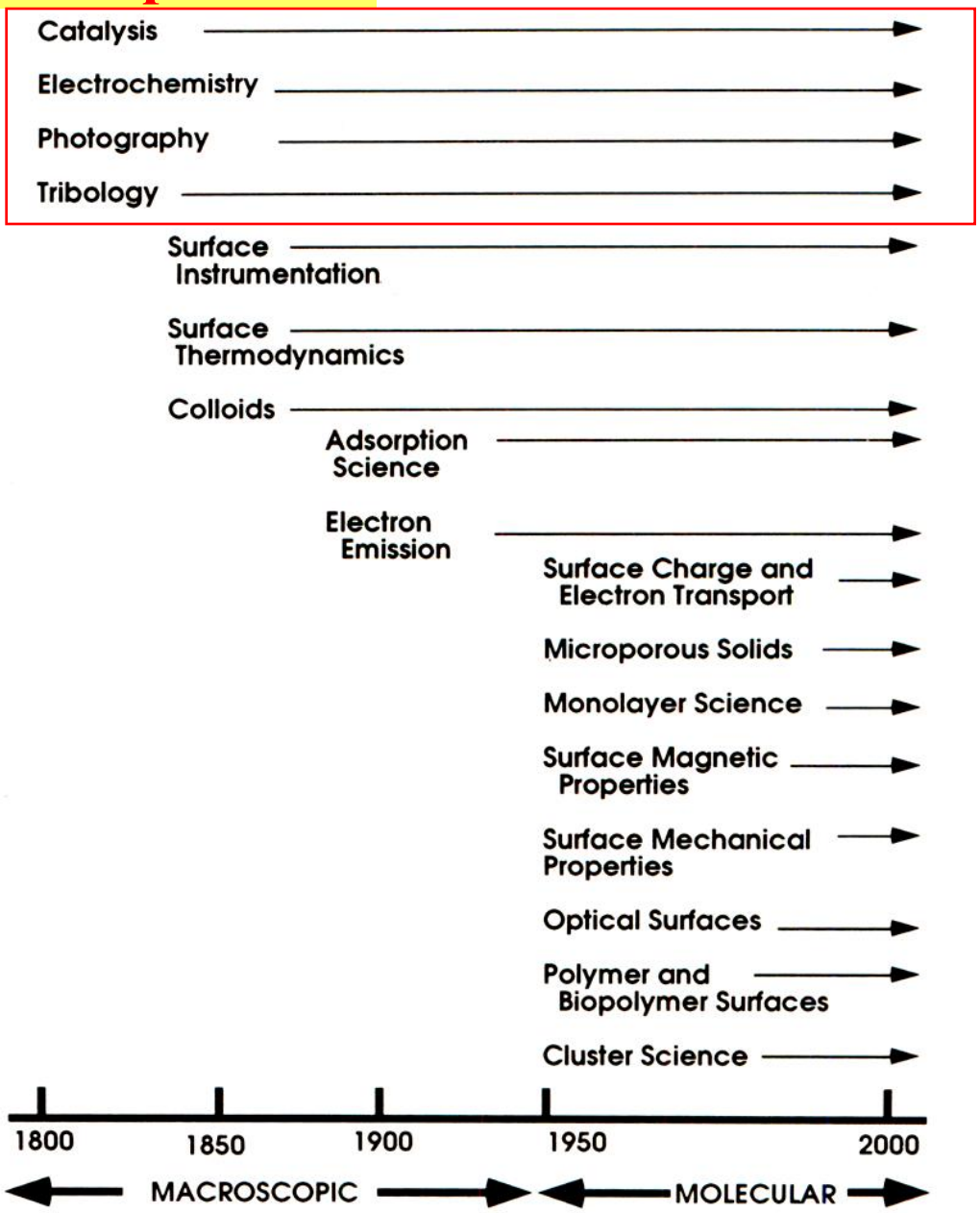


Figure 1.1. Timeline of the historical development of surface chemistry.

J. J. Berzelius (1779~1848)







1823年J. W. Döbereiner用Pt表面催化H_2和O_2的反应

（较早的表面技术）

Berzelius first used the “catalysis” in 1836

认为是一未知的力 (catalytic force)

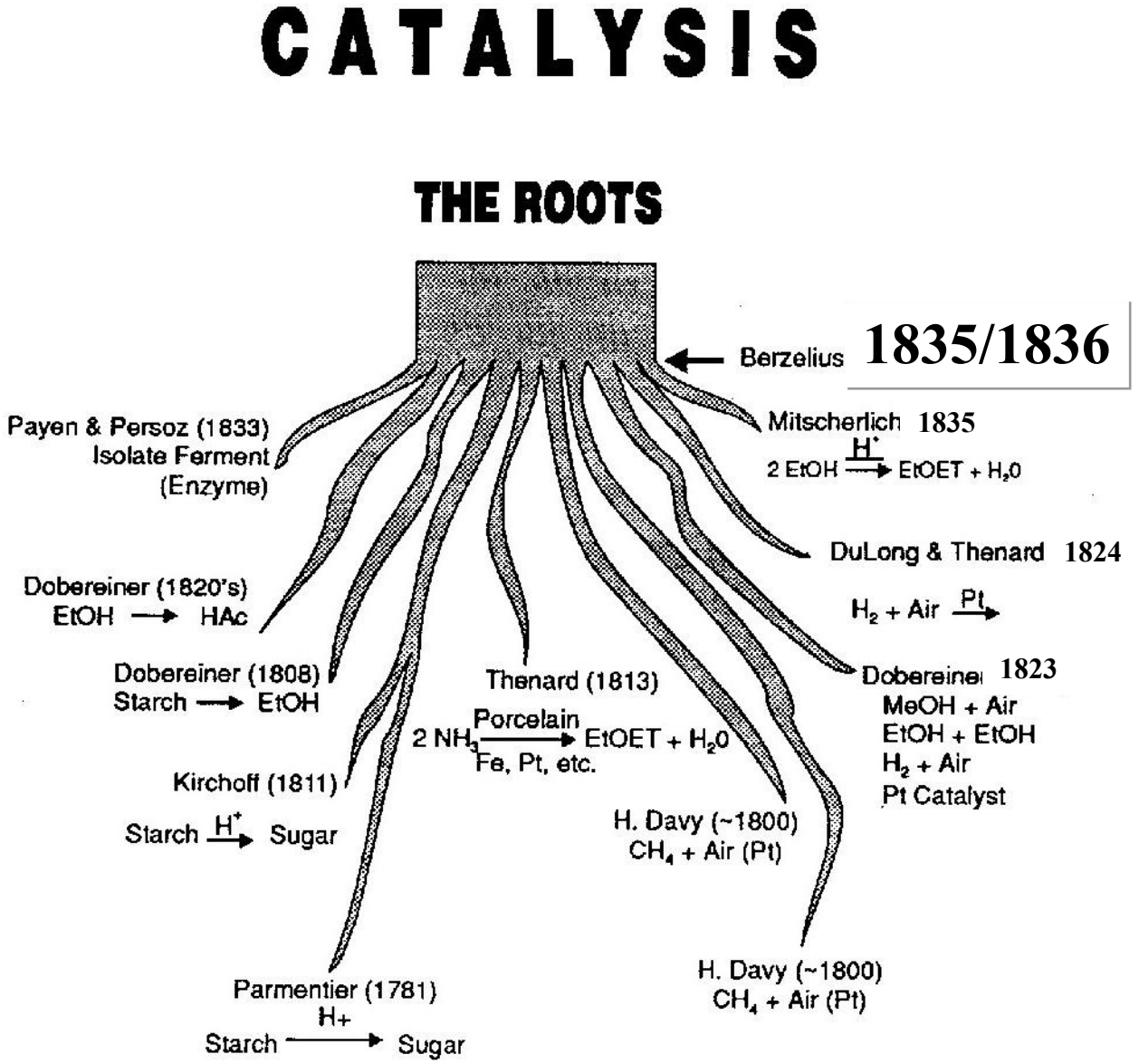
Wilhelm Ostwald (1853~1932)



Ostwald (1902):
A substance that changes the velocity of a reaction without itself being changed by the process.

Ostwald (1909 Nobel Prize) 比较清晰地提出催化作用的概念，并预言催化将在化学的各领域得到广泛认知，在工业中得到广泛应用

Catalysis in old days



Schematic representation of some of the studies that serve as a background for the formulation of the concept of catalysis

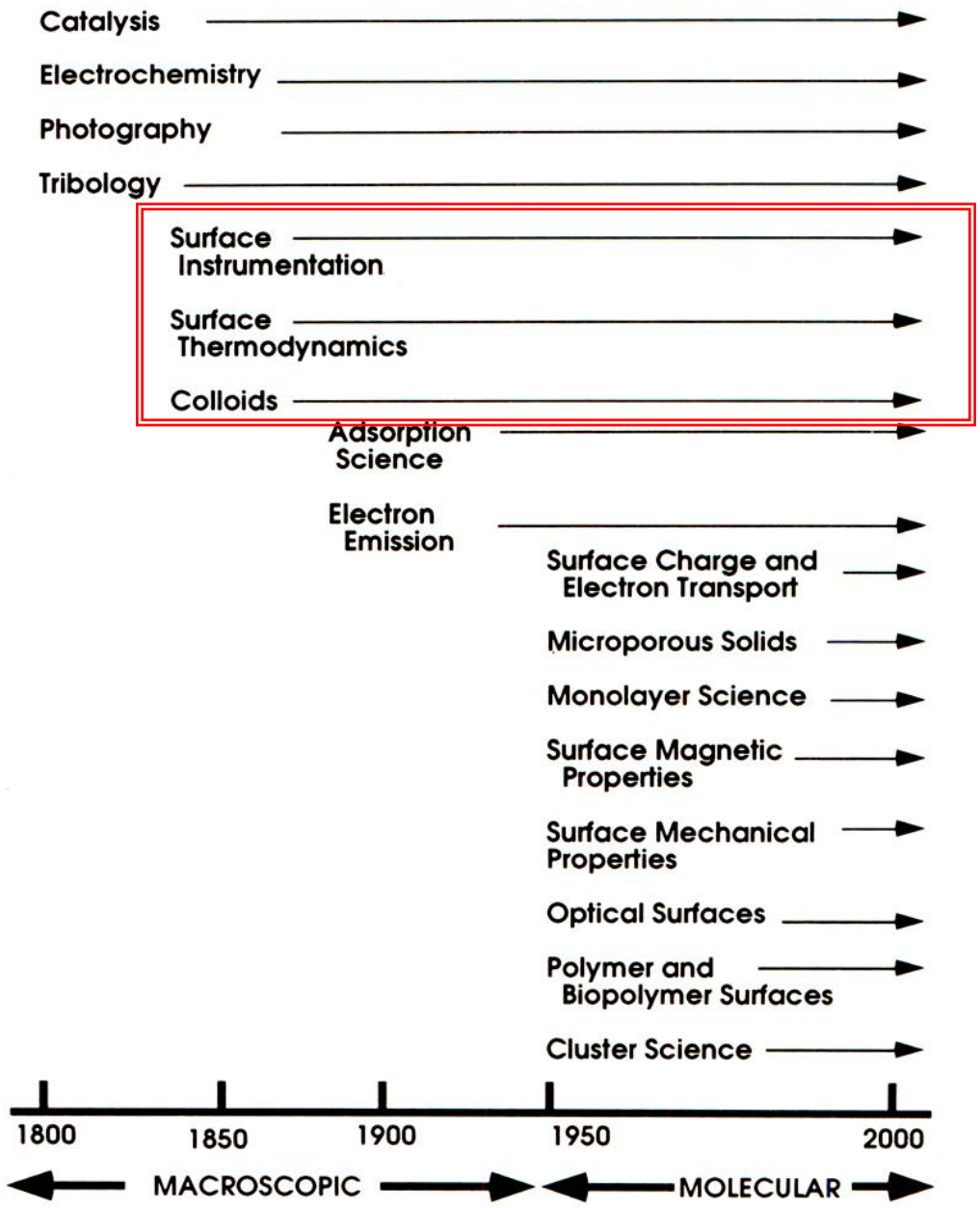
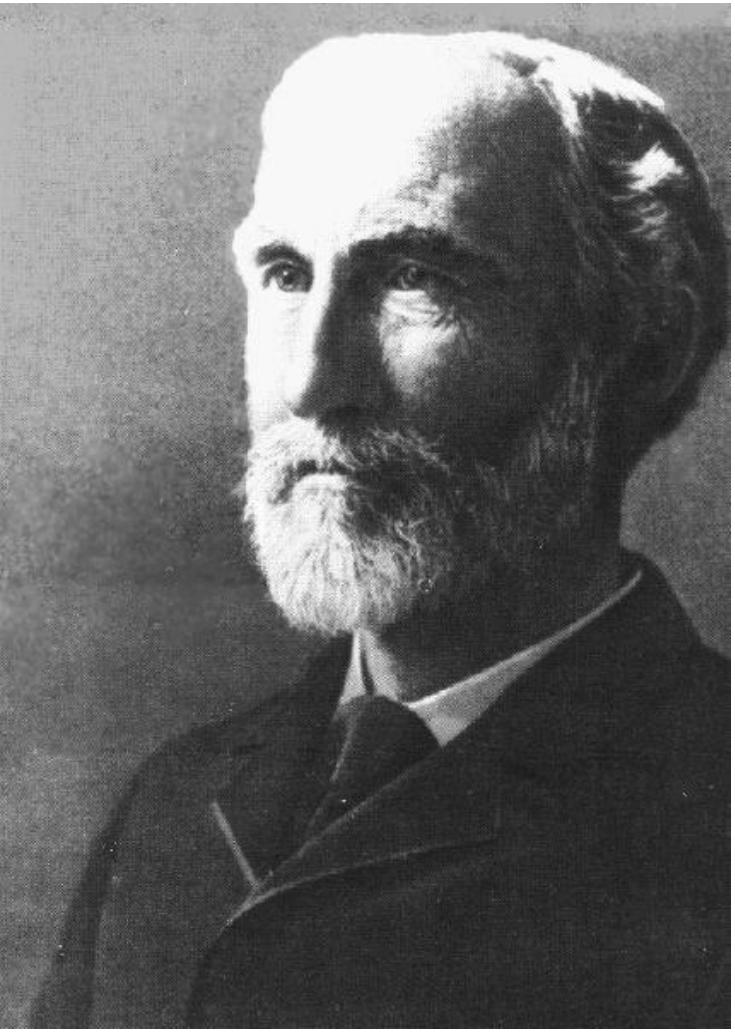


Figure 1.1. Timeline of the historical development of surface chemistry.

Josiah Willard Gibbs, 1839-1903年



化学热力学和统计热力学的奠基人

Ostwald称赞Gibbs:从内容到形式，他赋予物理化学整整一百年

Gibbs, 1877年提出适用于表面相的热力学理论

康乃狄格(Connecticut)科学院院报:
《论非均相物质之平衡》 (323页）

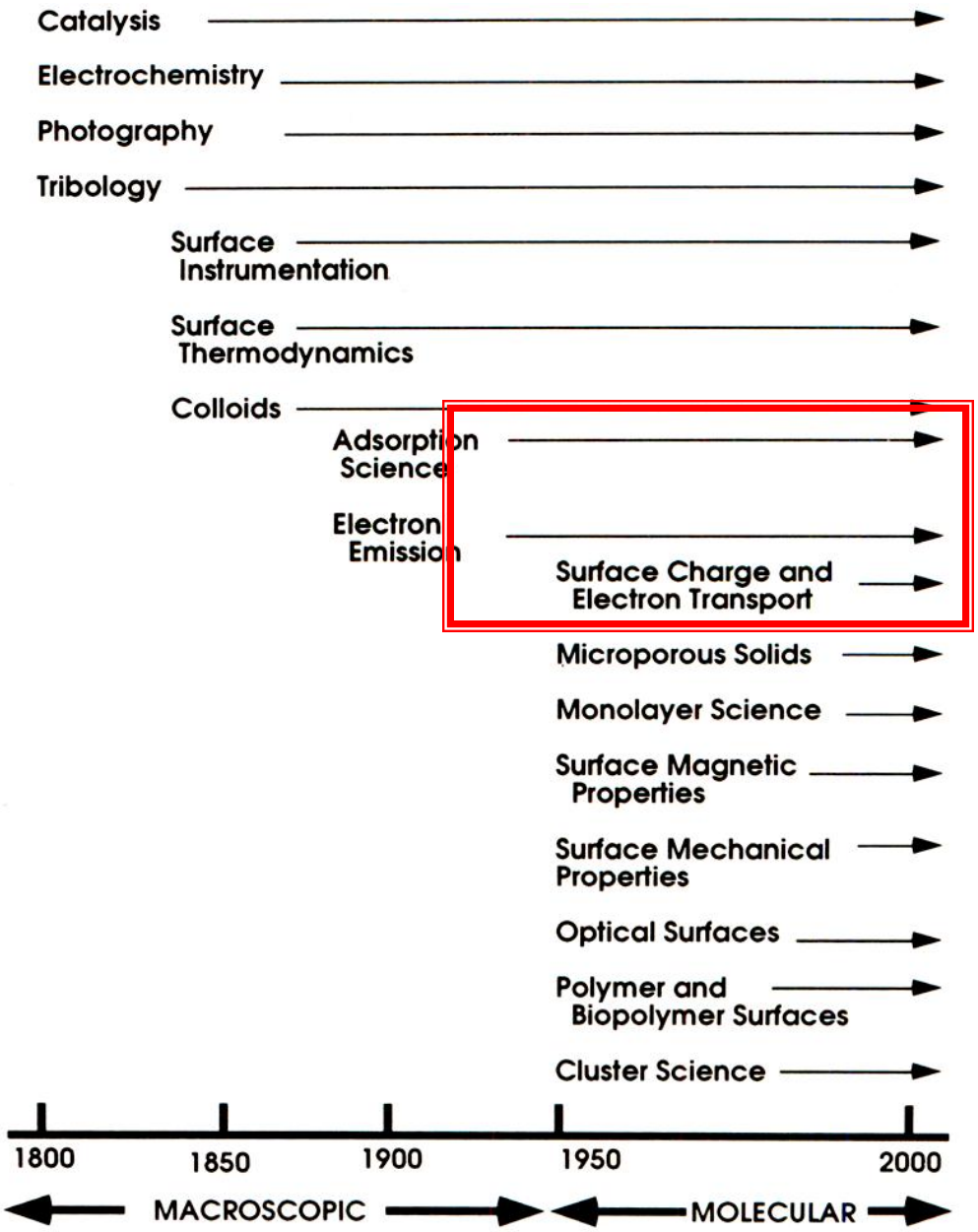
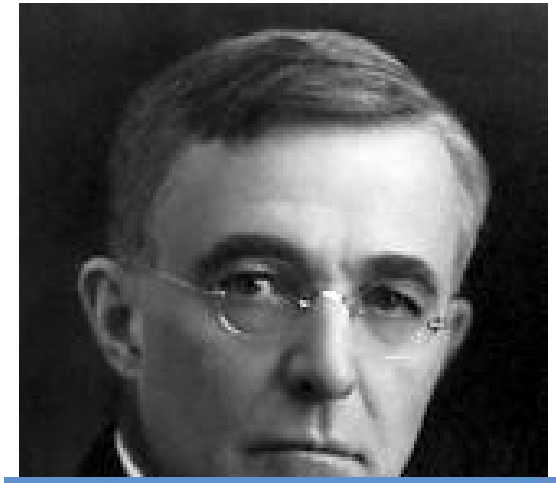


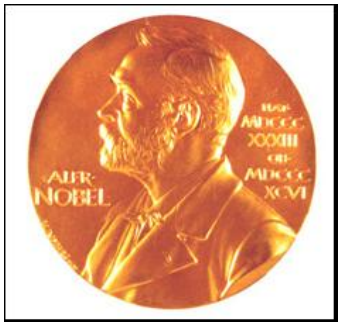
Figure 1.1. Timeline of the historical development of surface chemistry.



Irving Langmuir, 1881-1957

朗缪尔:空中有朵雨做的云

Nobel Prize (1932)





它的主要贡献是：（1）发明了充惰性气体的灯泡，延长了灯泡寿命；（2）发明了人工降雨；（3）提出了单分子层吸附理论，设计出研究表面张力的“表面天平”；（4）首创了表面化学和等离子体物理学等新学科；（5）发明了高真空计和水银扩散泵等。







第五次索尔维会议(Solvay Conference) (1927.10)

朗缪尔:空中有朵雨做的云



第三排：奥古斯特·皮卡尔德、亨里奥特、保罗·埃伦费斯特、爱德华·赫尔岑、顿德尔（en:Théophile de Donder）、埃尔温·薛定谔、维夏菲尔特（en:E.Verschaffelt）、沃尔夫冈·泡利、维尔纳·海森堡、拉尔夫·福勒、里昂·布里渊；第二排：彼得·德拜、马丁·努森、威廉·劳伦斯·布拉格、亨德里克·安东尼·克雷默、保罗·狄拉克、阿瑟·康普顿、路易·德布罗意、马克斯·玻恩、尼尔斯·玻尔；第一排：伊文·朗缪尔、马克斯·普朗克、玛丽·居里、亨德里克·洛伦兹、阿尔伯特·爱因斯坦、保罗·朗之万、查尔斯·欧仁·古耶、查尔斯·威耳逊、欧文·理查森

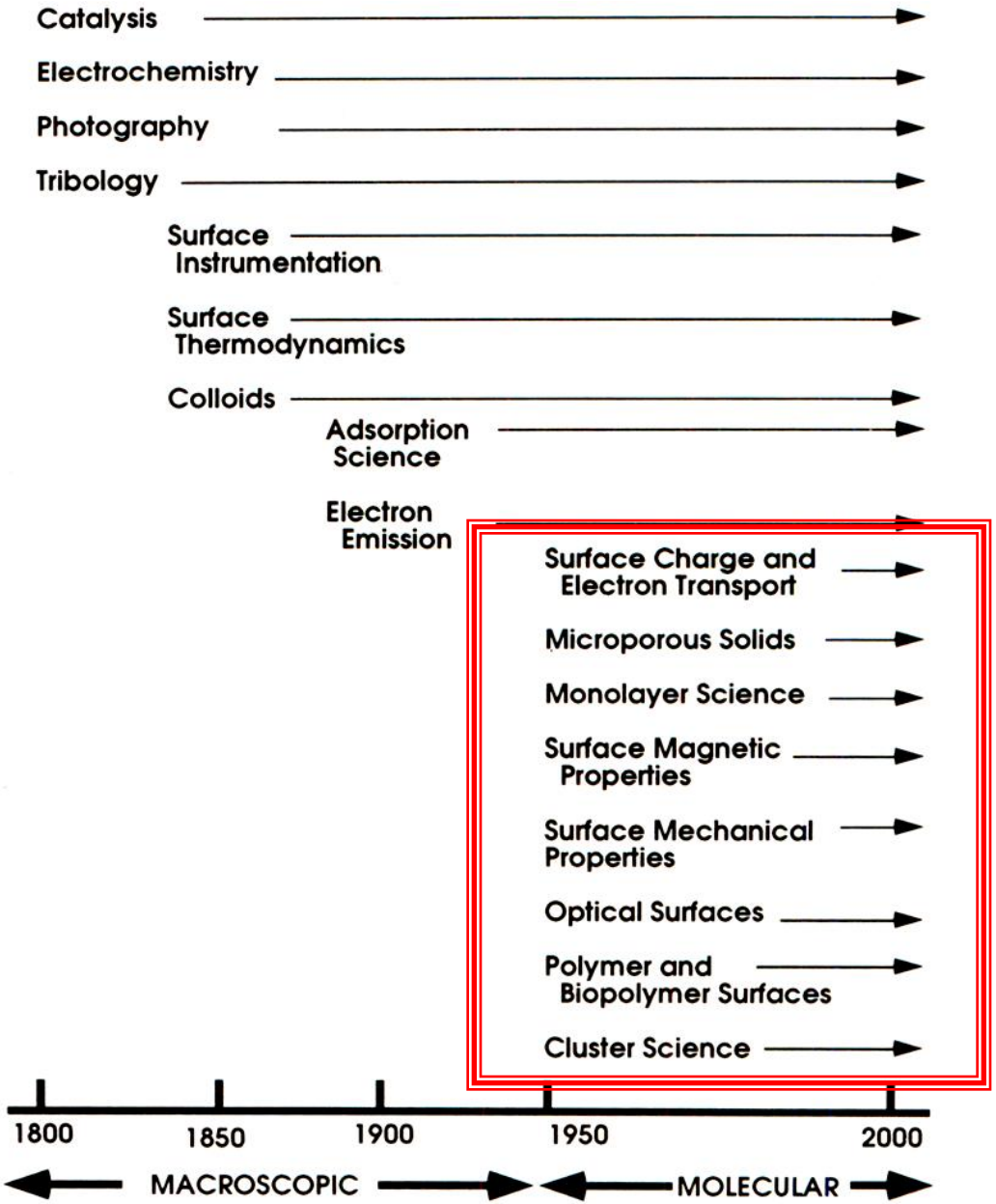
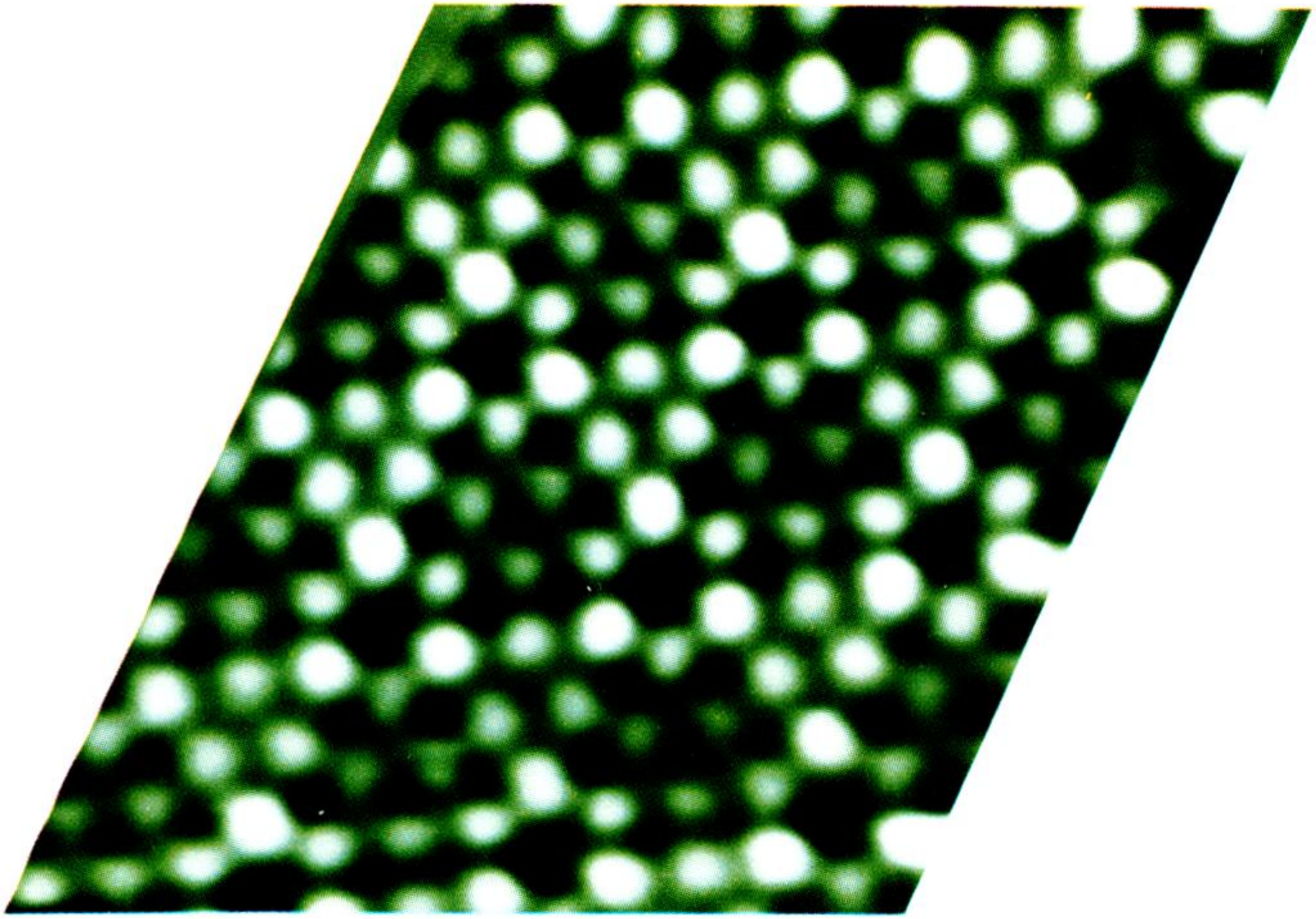
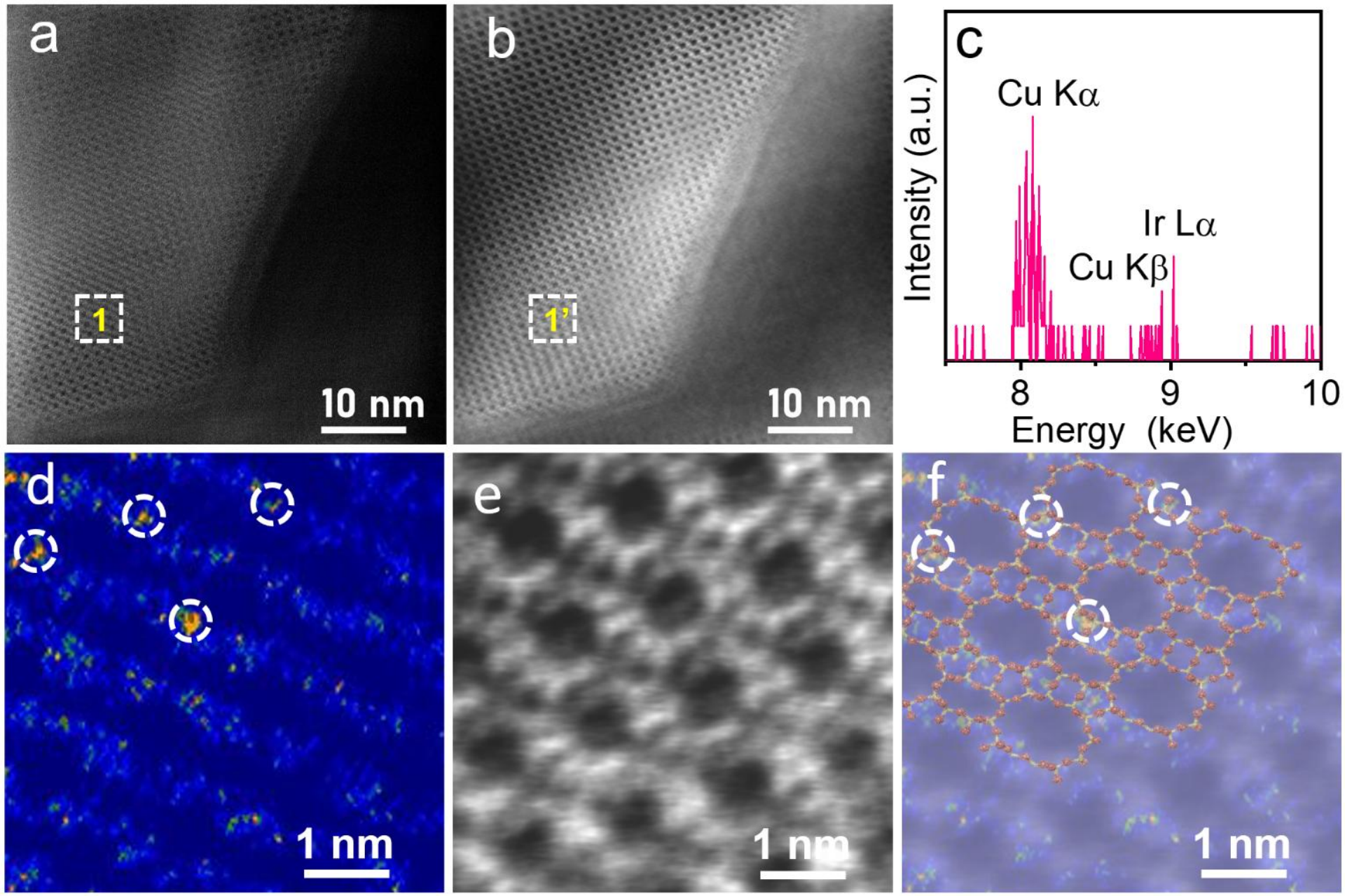
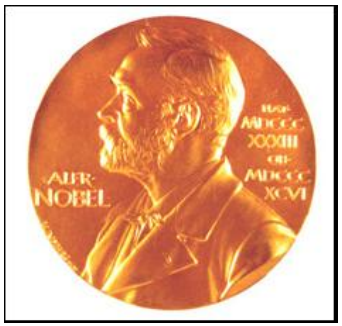


Figure 1.1. Timeline of the historical development of surface chemistry.



电镜下观测到的半导体硅表面的硅原子

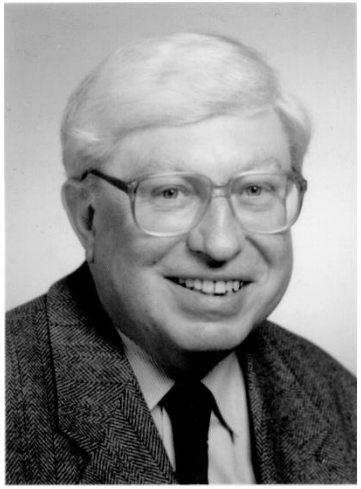

a
1
10 nm
b
1'
10 nm
c
Cu Kα
Ir Lα
Cu Kβ
Intensity (a.u.)
8
9
10
Energy (keV)
d
1 nm
e
1 nm
f
1 nm



The Nobel Prize in Chemistry 2007

"for his studies of chemical processes on solid surfaces"





Prof. Dr. Gerhard Ertl

Dept. of Physical Chemistry,
Fritz-Haber-Institut der
Max-Planck-Gesellschaft, Berlin

b.1936

"The father of modern surface chemistry"

National Medal of Science (2002)



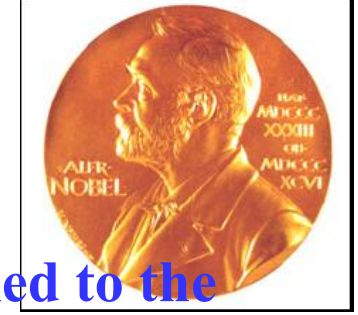
Prof. Gabor A. Somorjai

University of California,
Berkeley

http://www.berkeley.edu/news/media/releases/2002/05/10_somorjai.html

http://nobelprize.org/nobel_prizes/chemistry/laureates/2007/sci.html

Scientific Background on the Nobel Prize in Chemistry 2007
Chemical Processes on Solid Surfaces

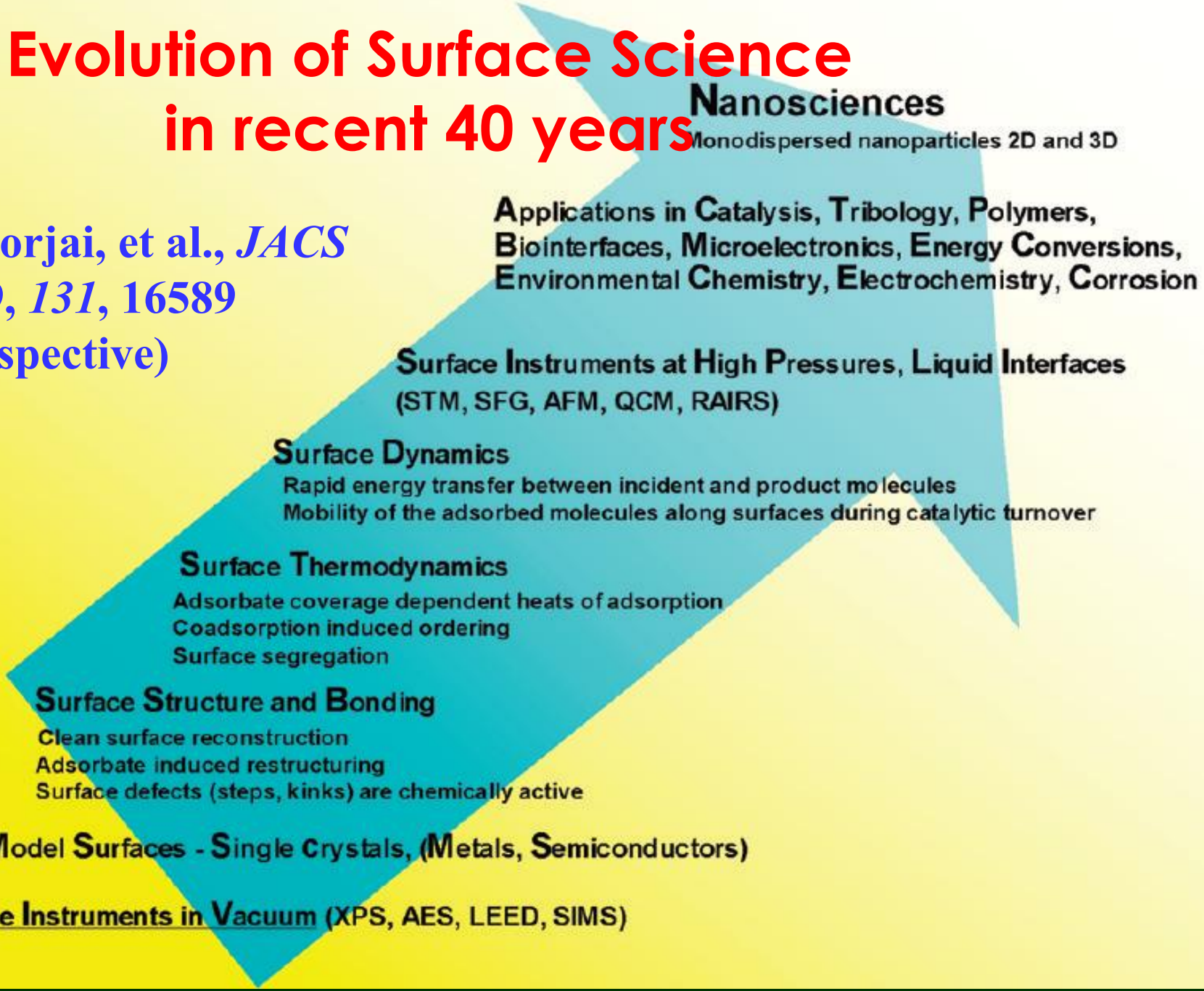


Ertl has developed a general methodology that can be applied to the important problems in molecular surface science. He has applied the methodology to some of the most central previously unanswered questions concerning molecules on surfaces. The investigations have been carried out with the greatest elegance in the experimental approach. His work is characterized by the ambition to always use the method best suited to solve the problem at hand. Ertl is never satisfied with an isolated interesting observation. Instead the studies are brought to their logical conclusions. Through his accurate studies he has provided a firm basis for our thinking about molecular processes at surfaces.

Håkan Wennerström
Professor of Theoretical Physical Chemistry, Lund University
Member of the Nobel Committee for Chemistry
Sven Lidin
Professor of Inorganic Chemistry, Stockholm University
Member of the Nobel Committee for Chemistry

Evolution of Surface Science in recent 40 years

Somorjai, et al., *JACS* 2009, *131*, 16589 (Perspective)



1.2 Concepts related to surfaces-1

1. Surface concentration (density)

体相密度

bulk: 1 g/cm^3～分子密度$\rho \approx 5\times10^{22}$ molecules/cm^3

表面密度

分子密度$\sigma \approx \rho^{2/3} = 1\times10^{15}$ molecules/cm^2

根据所研究面的不同可能会相差1-2个数量级

例子

	M(g/mol)	ρ(g/cm^3)	σ(cm^{-2})
Hg	200	13.6	1.2×10^{14}
Cu	63.5	8.9	1.9×10^{15}
C_6H_6	78	0.879	3.6×10^{14}

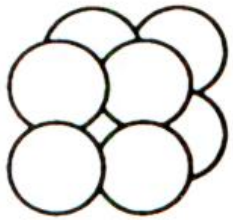
1.2 Concepts related to surfaces-2

2. Dispersion （分散度）

***D*= number of surface atoms /total number of atoms**

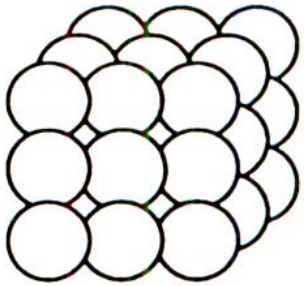
考虑简单立方堆积

n= 8



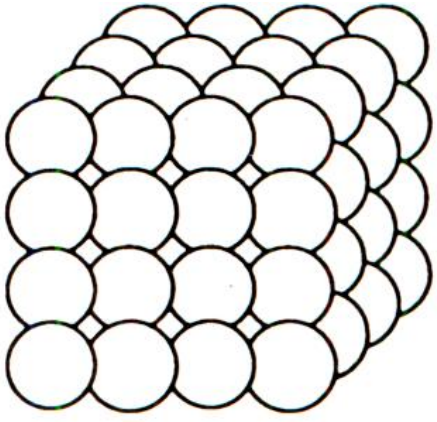
***D*=1**

n= 27



***D*= 26/27= 0.963**

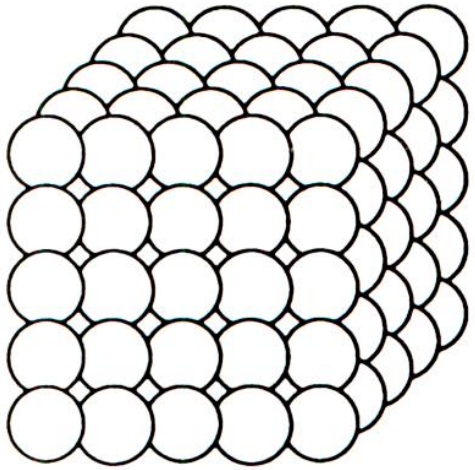
n= 64



***D*= 56/64= 0.875**

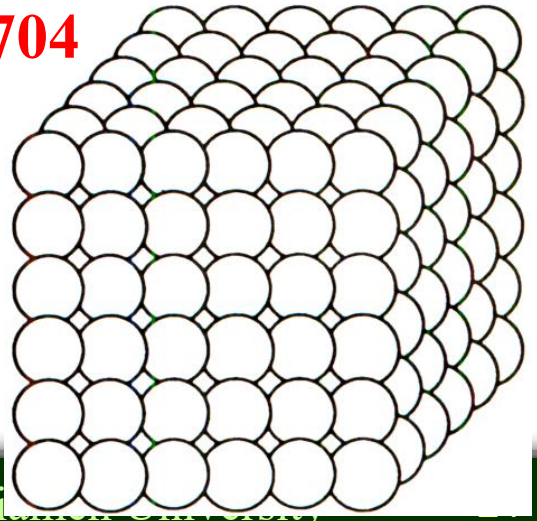
n=125

***D*= 98/125= 0.784**



n=216

***D*= 152/216= 0.704**



Dispersion and particle size

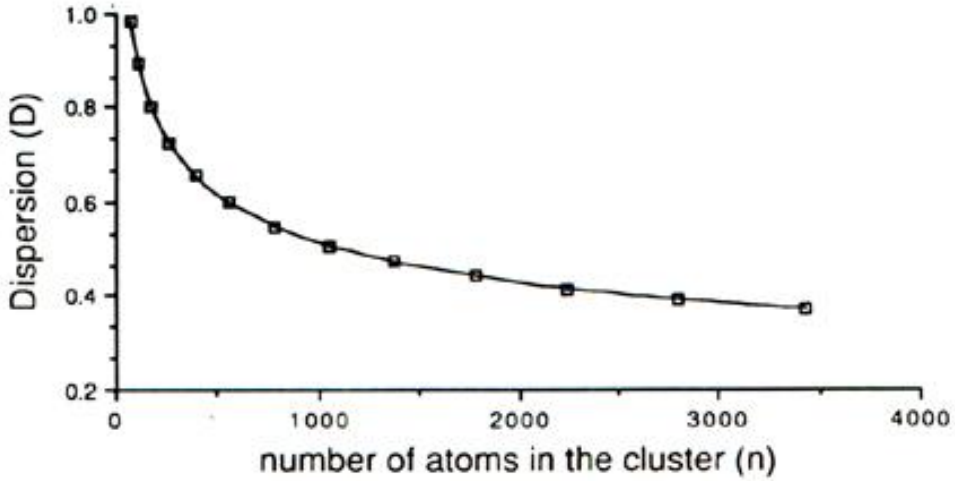
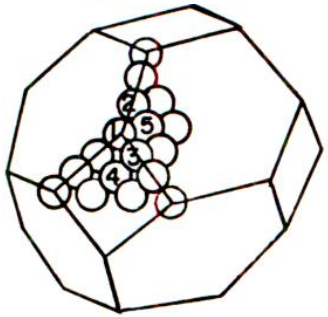


Figure 1.7. Clusters of atoms with single cubic packing having 8, 27, 64, 125, and 216 atoms. In an eight-atom cluster, all of the atoms are on the surface. However, the dispersion, $\mathfrak{D}$, defined as the number of surface atoms divided by the total number of atoms in the cluster, declines rapidly with increasing cluster size. this is shown in the lower part of the figure.



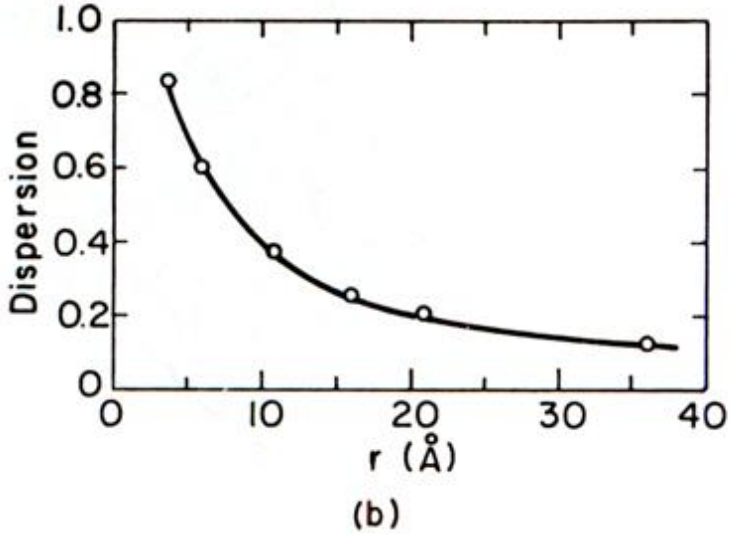
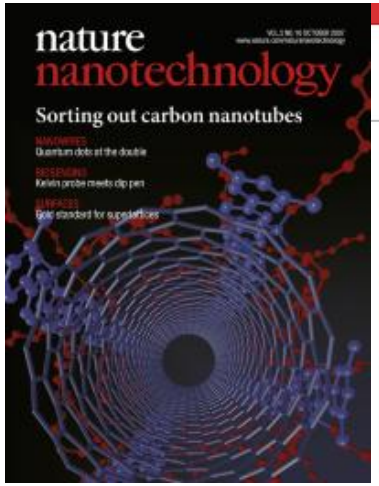


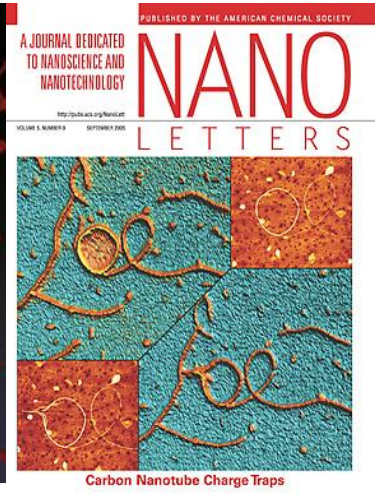
Figure 1.8. Dispersion as a function of size for particles with cubo-octahedral packing.

Nano Science & Surface Science

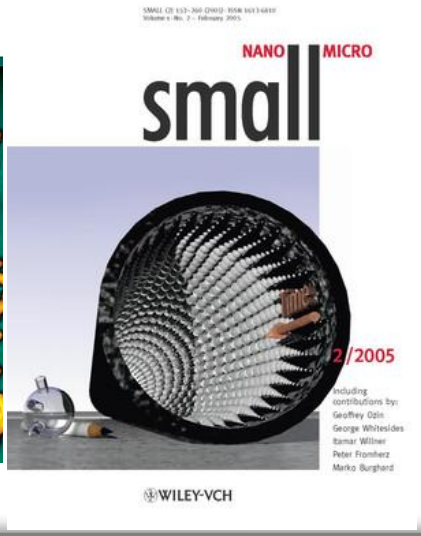
随粒子尺寸减小，分散度迅速增大，表面效应十分显著。

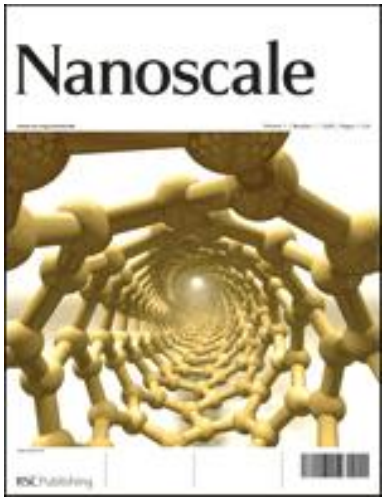
包括多相催化在内的许多表面过程仅与表面原子有关，因而这些体系中常常有必要减小活性粒子大小，增加分散度。





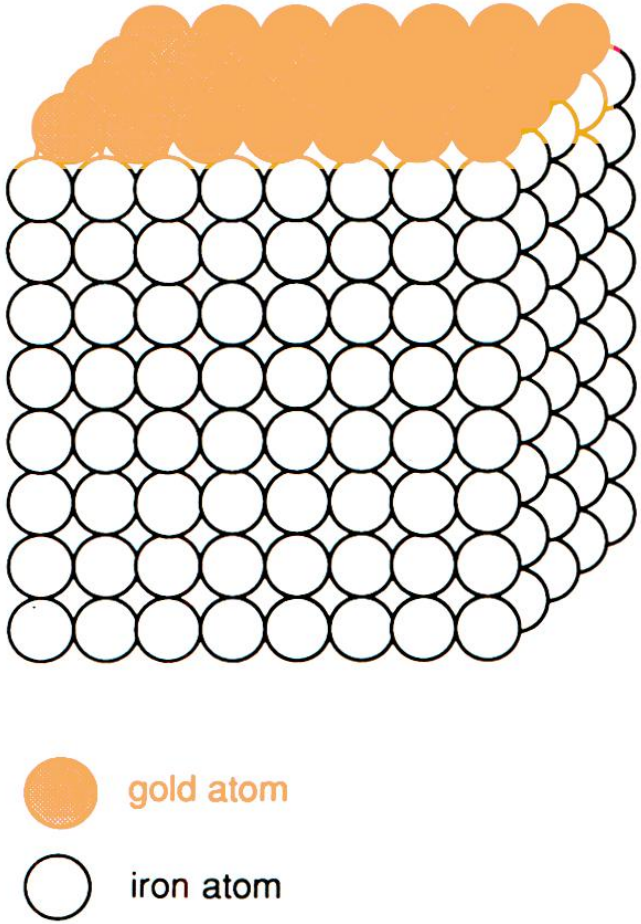




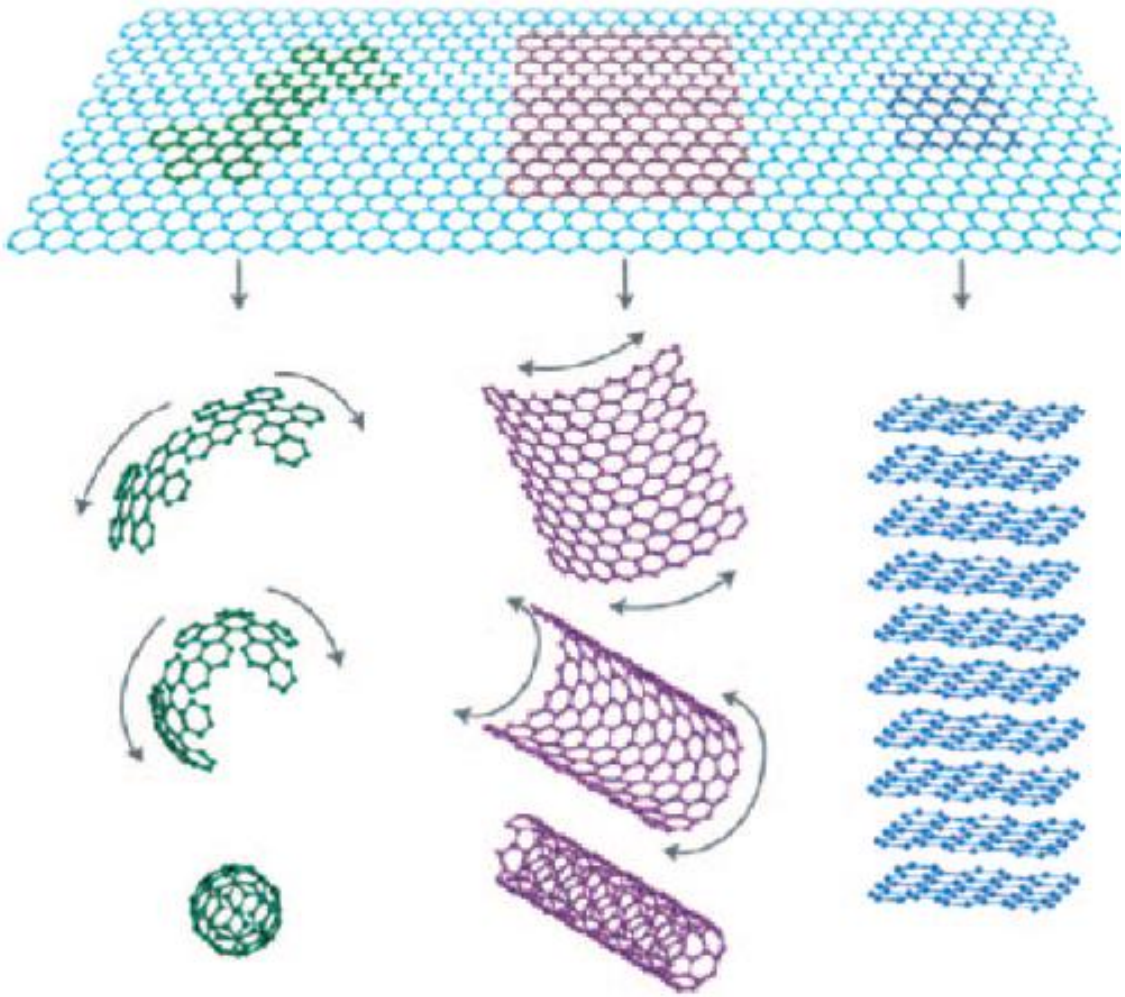


3. Thin films-2D materials

Thin layer An iron particle with one surface covered with a monolayer of gold atoms. When it comes to surface properties such as adsorption or catalysis, one monolayer of atoms is all that is needed to carry out the necessary chemistry. Because of the very few atoms needed, such a device is inexpensive. For other applications a thin film of up to 50 layers is needed. For this reason, thin films are very often utilized to carry out surface reactions (adsorption, catalysis) and to impart mechanical properties (hardness, lubrication), optical properties (reflectivity, color), and electrical properties (conductivity, charge retention).



New carbon nanomaterials



A. K. Geim, K. S. Novoselov, *Nat. Mater.* **2007**, *6*, 183

1.2 Concepts related to surfaces-4

4. Internal Surface

Porous solid materials

孔	大小	例
大孔(macropore)	>50 nm	多孔玻璃 煤炭
介孔(mesopore)	2～50 nm	气凝胶 (> 10nm) 层状粘土(1-10nm) 介孔分子筛 (MCM-41, SBA-15) (2-30nm)
微孔(micropore)	<2nm	沸石分子筛 (<1.4 nm) MOF材料 活性炭（0.6nm）

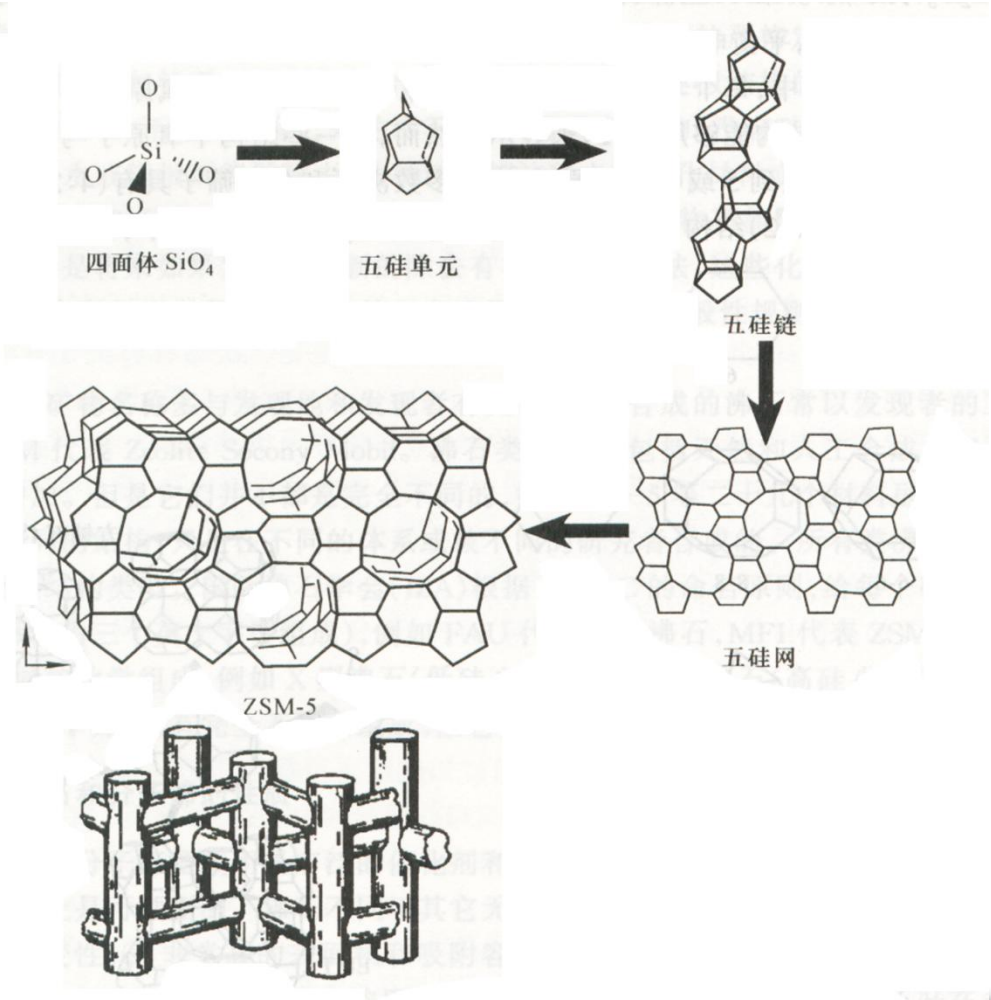
IUPAC: 关于孔的定义

1.2 Concepts related to surfaces-4

4. Internal Surface

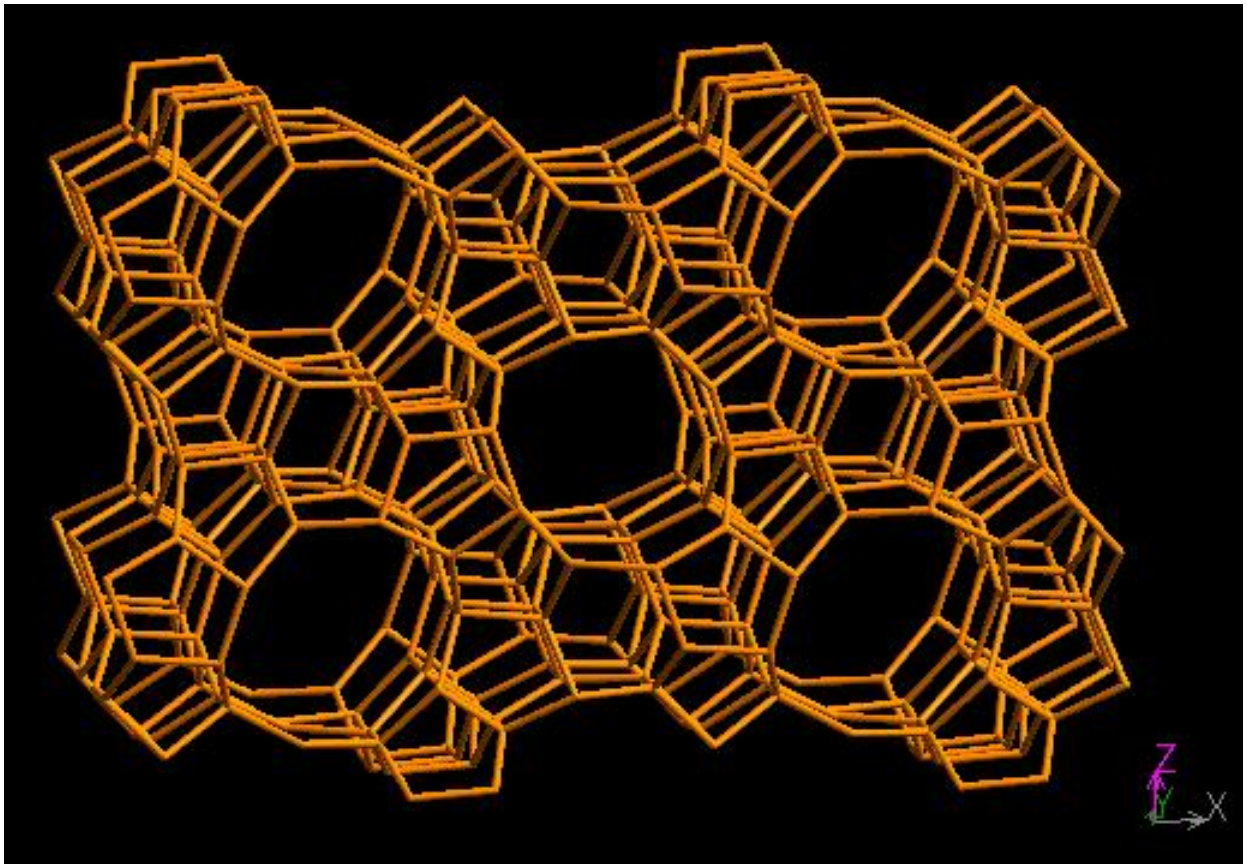
规则孔道材料

(1) 微孔分子筛：ZSM-5, Y, MCM-22



ZSM-5结构

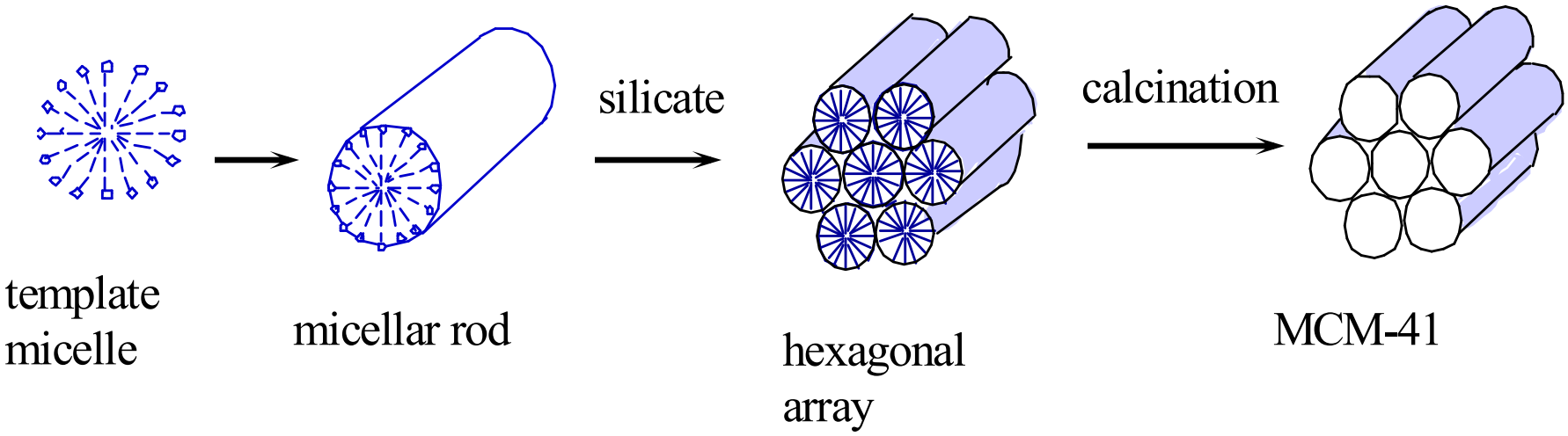
ZSM-5 (Zeolite Socony Mobil 5)
沸石分子筛



1.2 Concepts related to surfaces-4

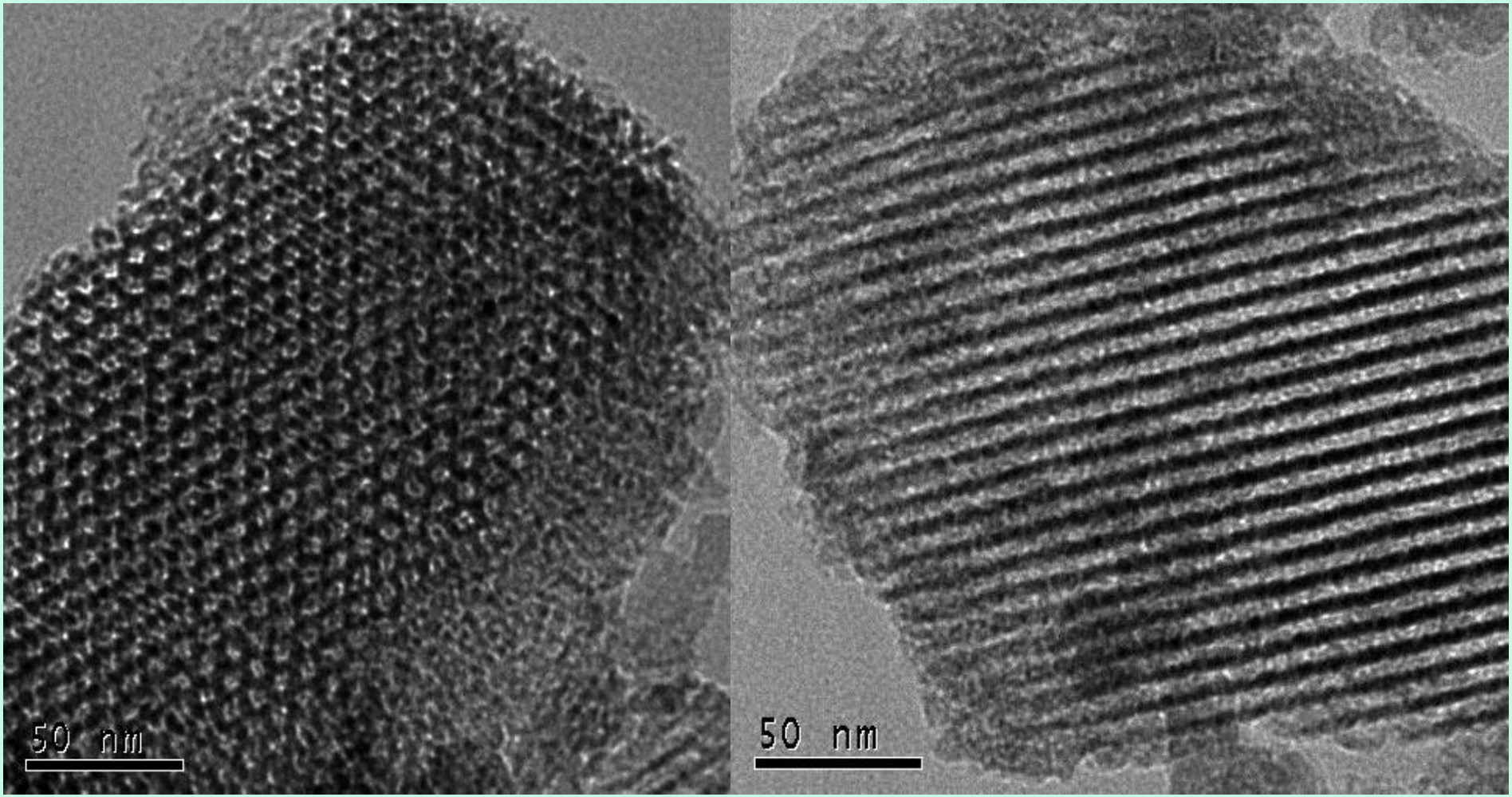
4. Internal Surface 规则孔道材料

(2) 介孔分子筛: MCM-41, MCM-48, SBA-15等



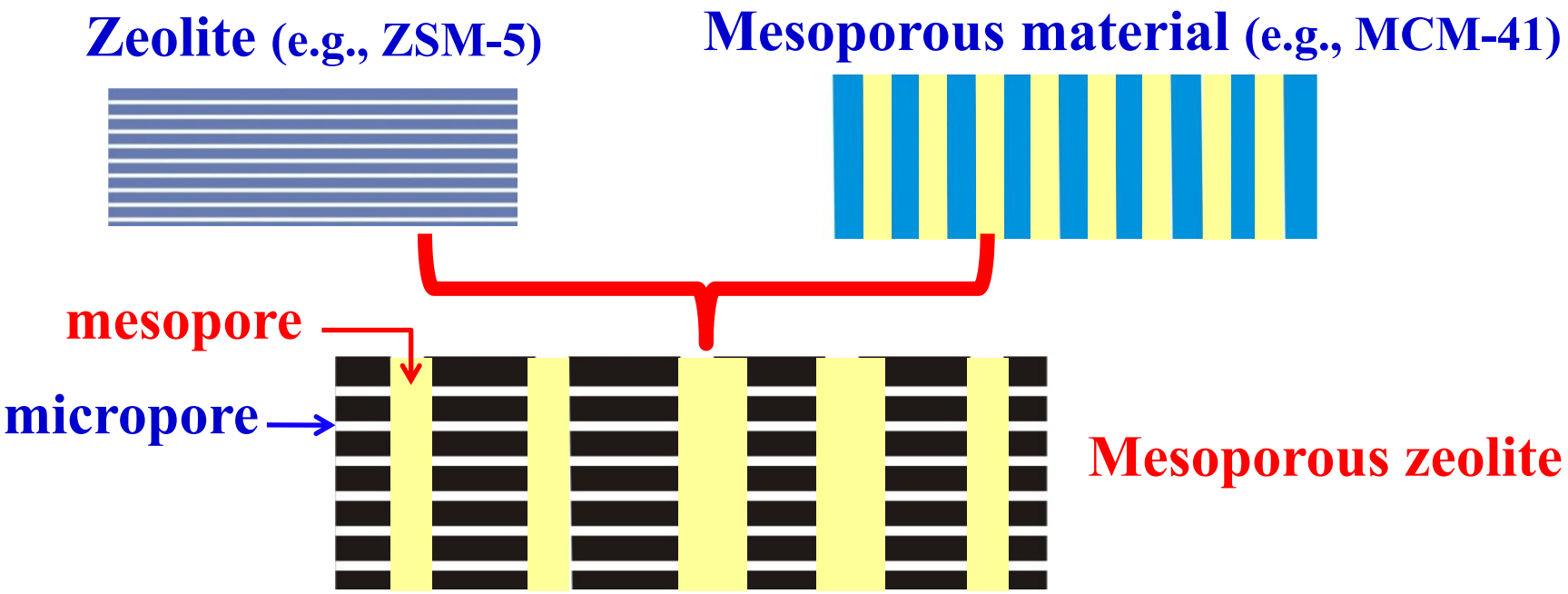
MCM-41 (Mobil composition of matter-41)

形成及结构



TEM of SBA-15 (Santa Barbara Amorphous-15)

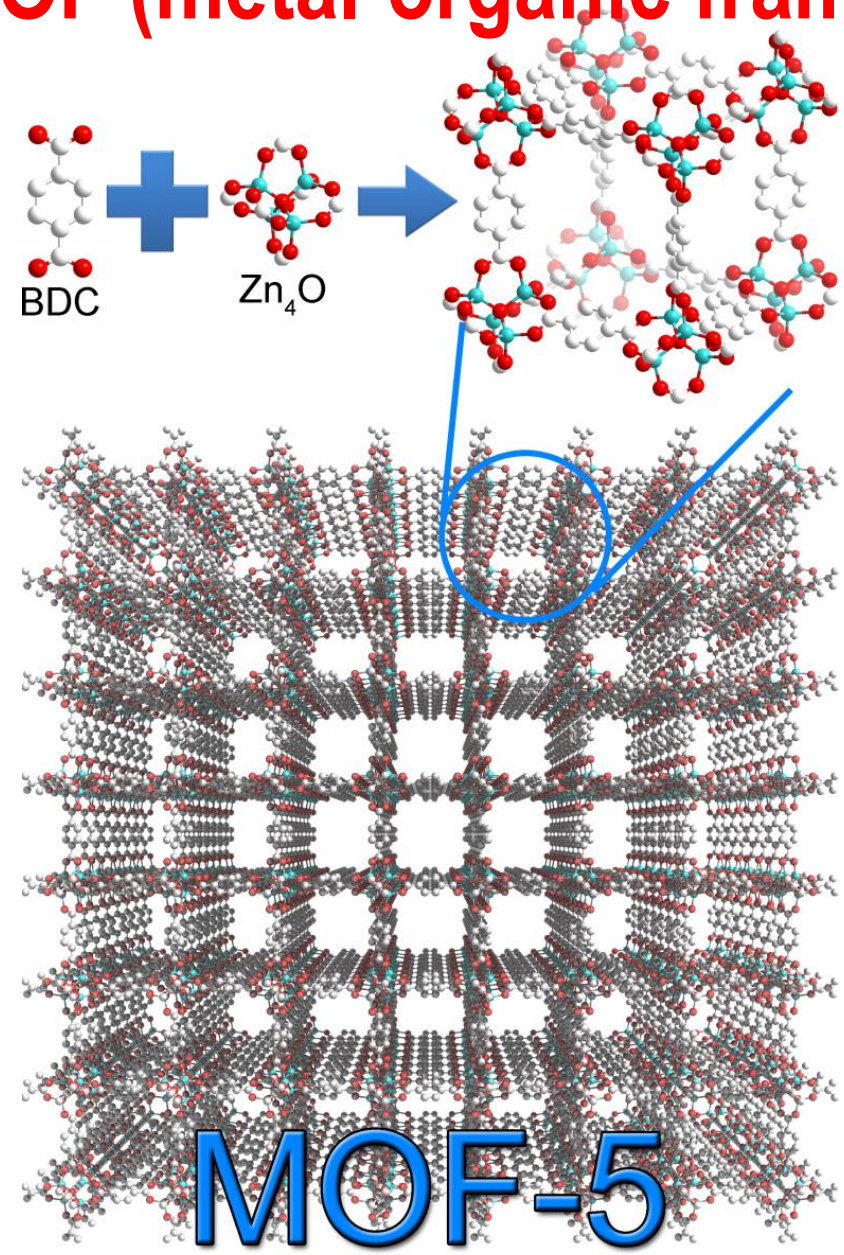
Hierarchical porous materials



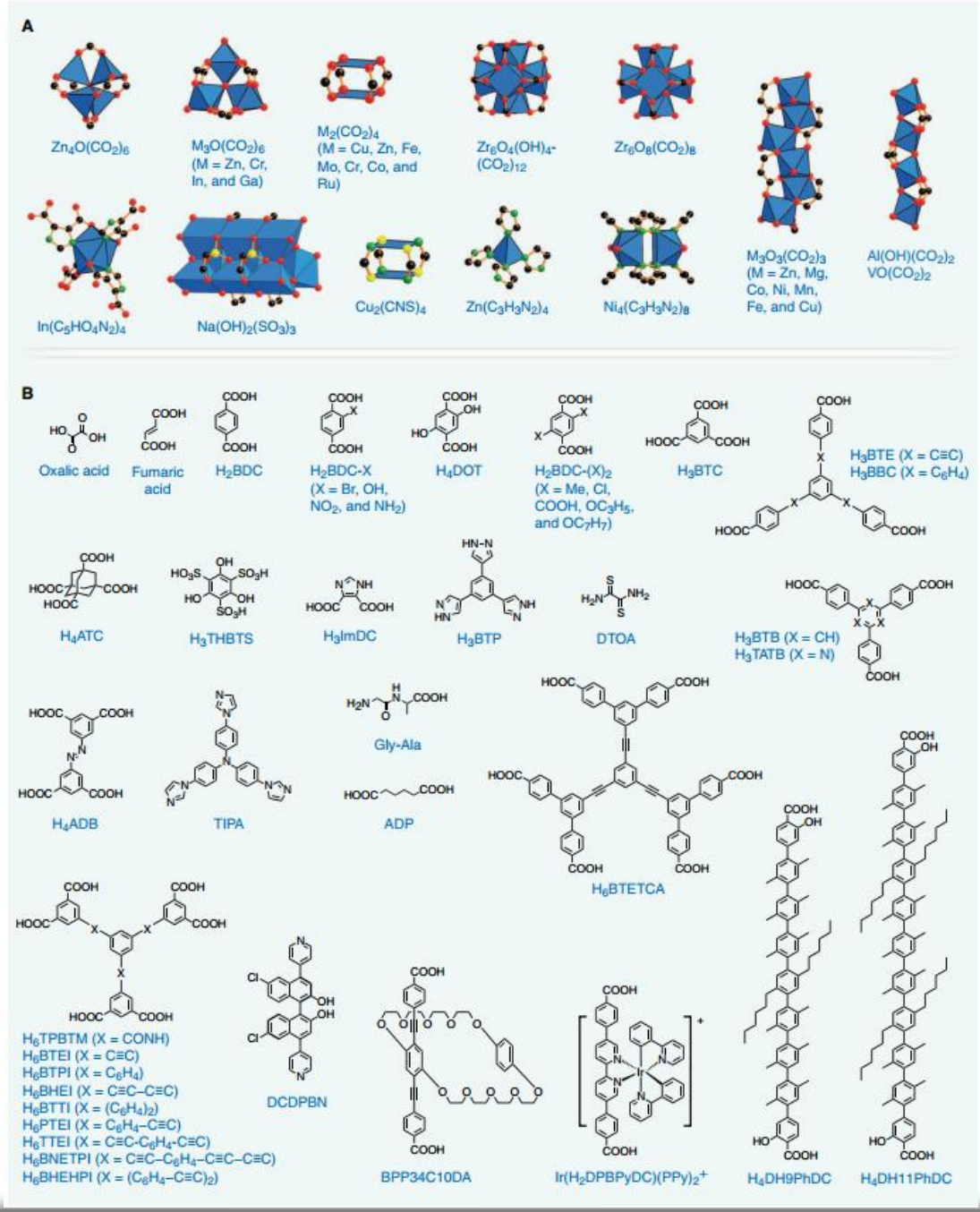
Mesoporous zeolites combine the advantages of **conventional zeolites** and **mesoporous materials** (efficient mass-transportation). **Mesoporous zeolites have attracted much attention as a new type of materials.**

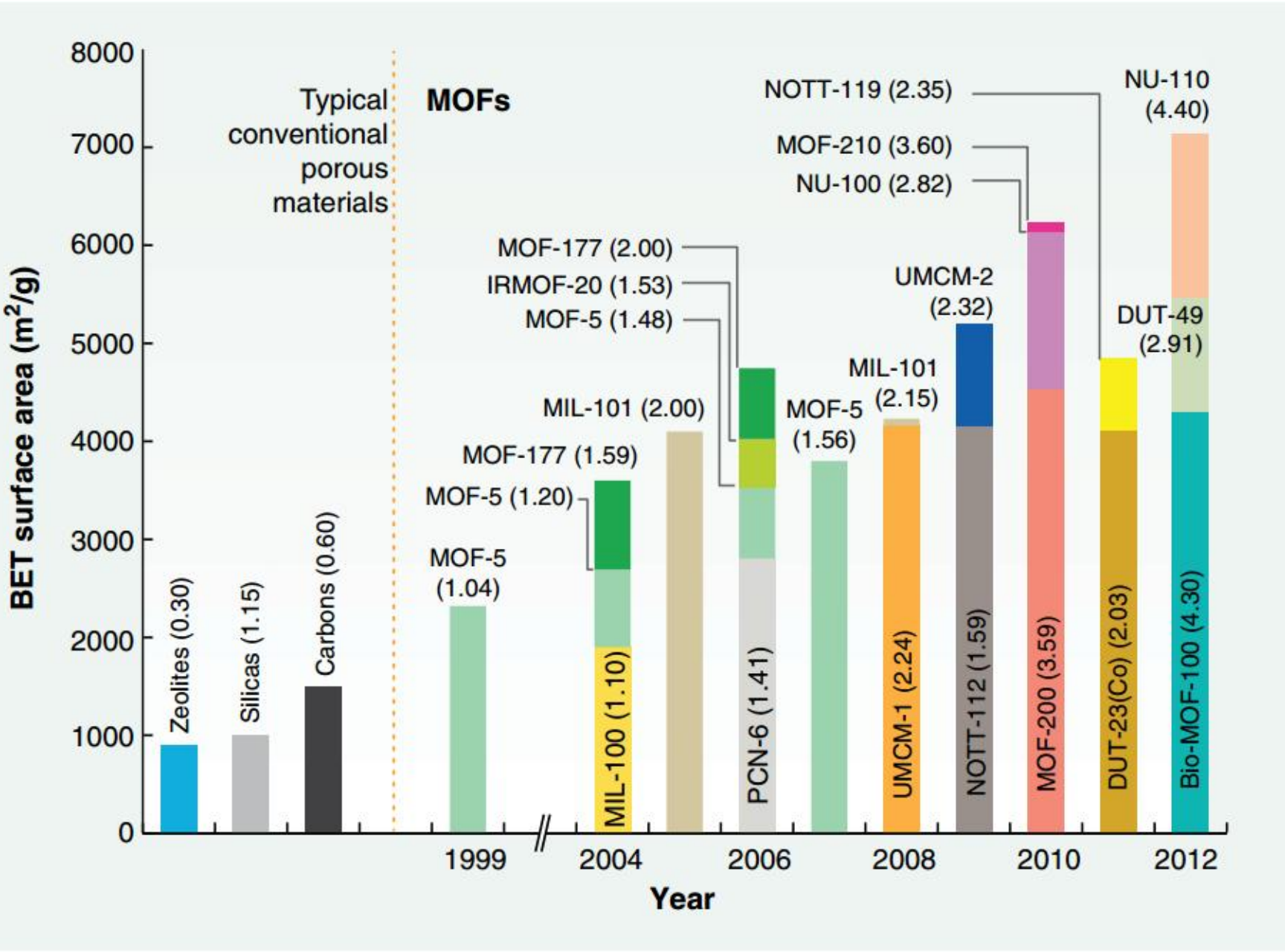
Hartman, *Angew. Chem. Int. Ed.* 2004, *43*, 5880

MOF (metal-organic framework)



7000 m^2 g^{-1}


A
$Zn_4O(CO_2)_6$
$M_3O(CO_2)_6$ (M = Zn, Cr, In, and Ga)
$M_2(CO_2)_4$ (M = Cu, Zn, Fe, Mo, Cr, Co, and Ru)
$Zr_6O_4(OH)_4(CO_2)_{12}$
$Zr_6O_8(CO_2)_8$
$M_3O_3(CO_2)_3$ (M = Zn, Mg, Co, Ni, Mn, Fe, and Cu)
$Al(OH)(CO_2)_2$
$VO(CO_2)_2$
$In(C_5HO_4N_2)_4$
$Na(OH)_2(SO_3)_3$
$Cu_2(CNS)_4$
$Zn(C_3H_3N_2)_4$
$Ni_4(C_3H_3N_2)_8$
B
Oxalic acid
Fumaric acid
H_2BDC
H_2BDC-X (X = Br, OH, NO_2, and NH_2)
H_4DOT
H_2BDC-$(X)_2$ (X = Me, Cl, COOH, OC_3H_5, and OC_7H_7)
H_3BTC
H_3BTE (X = C≡C)
H_3BBC (X = C_6H_4)
H_4ATC
H_3THBTS
H_3ImDC
H_3BTP
DTOA
H_3BTB (X = CH)
H_3TATB (X = N)
H_4ADB
TIPA
Gly-Ala
ADP
$H_6BTETCA$
H_6TPBTM (X = CONH)
H_6BTEI (X = C≡C)
H_6BTPI (X = C_6H_4)
H_6BHEI (X = C≡C–C≡C)
H_6BTTI (X = $(C_6H_4)_2$)
H_6PTEI (X = C_6H_4–C≡C)
H_6TTEI (X = C≡C-C_6H_4-C≡C)
$H_6BNETPI$ (X = C≡C–C_6H_4–C≡C–C≡C)
$H_6BHEHPI$ (X = $(C_6H_4$–C≡C$)_2$)
DCDPBN
BPP34C10DA
$Ir(H_2DPBPyDC)(PPy)_2^+$
$H_4DH9PhDC$
$H_4DH11PhDC$


BET surface area (m^2/g)
8000
7000
6000
5000
4000
3000
2000
1000
0
Typical conventional porous materials
MOFs
Zeolites (0.30)
Silicas (1.15)
Carbons (0.60)
MOF-5 (1.04)
MIL-100 (1.10)
MOF-177 (1.59)
MOF-5 (1.20)
MIL-101 (2.00)
MOF-177 (2.00)
IRMOF-20 (1.53)
MOF-5 (1.48)
PCN-6 (1.41)
MOF-5 (1.56)
UMCM-1 (2.24)
MIL-101 (2.15)
UMCM-2 (2.32)
NOTT-112 (1.59)
NOTT-119 (2.35)
MOF-210 (3.60)
NU-100 (2.82)
MOF-200 (3.59)
DUT-23(Co) (2.03)
NU-110 (4.40)
DUT-49 (2.91)
Bio-MOF-100 (4.30)
1999
2004
2006
2008
2010
2012
Year

5. Adsorption

当接近表面时，每一个原子或分子都会遇到一个引力使其同表面结合，这一过程称为**吸附 (adsorption)**。

吸附一般是放热过程，虽然按热力学放热过程的焓变应为负值，但一般ΔH_{ads}表示为正值。

吸附原子或分子在表面的停留时间可用吸附热估算，

$$\tau = \tau_0 \exp\left(\frac{\Delta H_{ads}}{RT}\right)$$

τ_0 是表面原子的振动时间10^{-11}～10^{-12} s

当ΔH_{ads}> 63 kJ mol^{-1} (15 kcal mol^{-1}), $\tau \approx 1$ s

吸附分子的表面浓度

$\sigma = F\tau$

F: 单位时间单位面积上流过的气体量(通量)

1.3 Unique features of solid surfaces

1 Surface heterogeneity

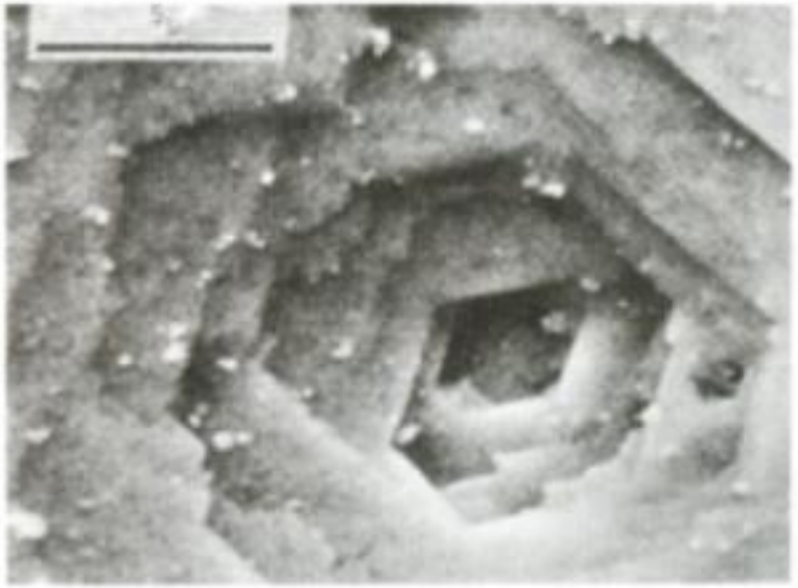
2 Clean surface and ultra high vacuum (UHV)

1. Surface heterogeneity

A. 表面并不平坦



SEM picture of Zn crystal surface



B. 更微观地看表面

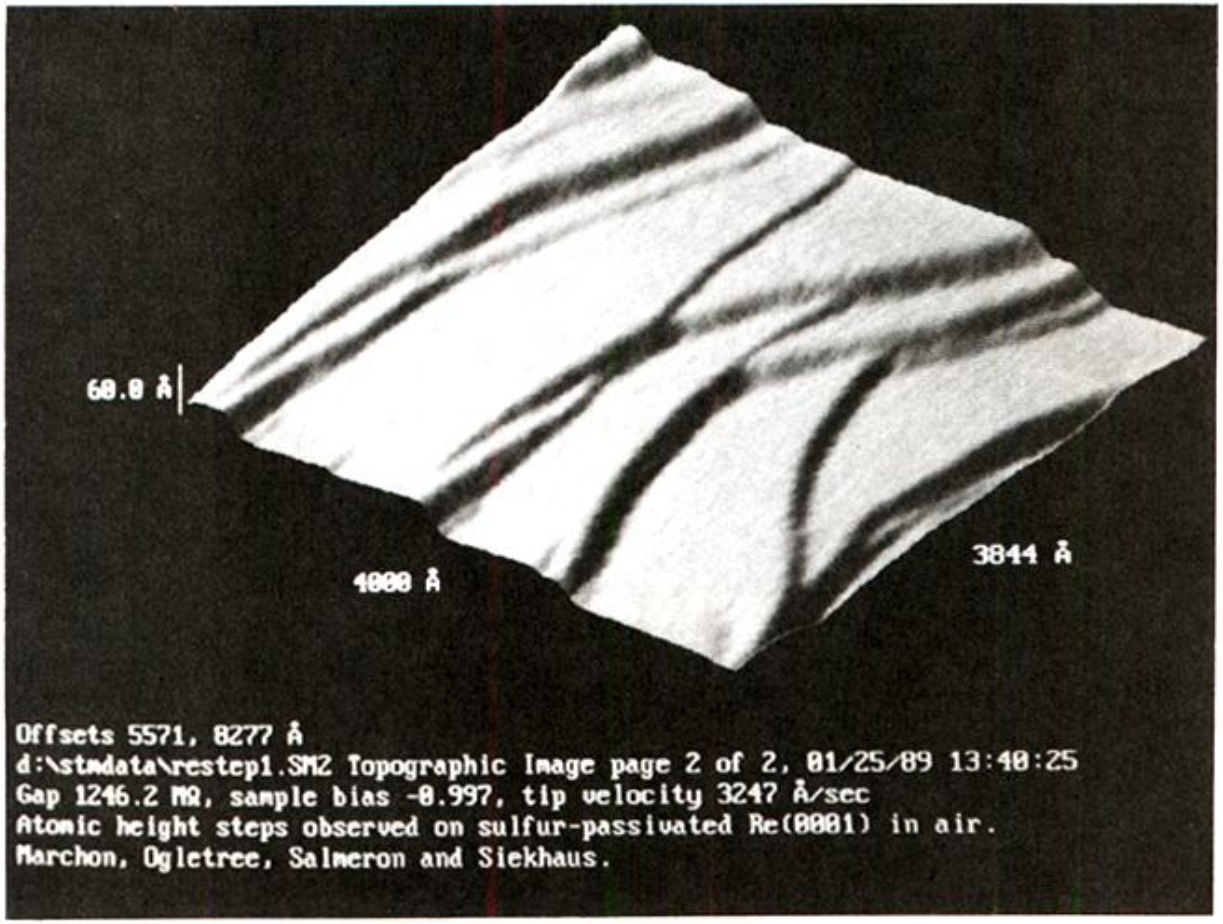
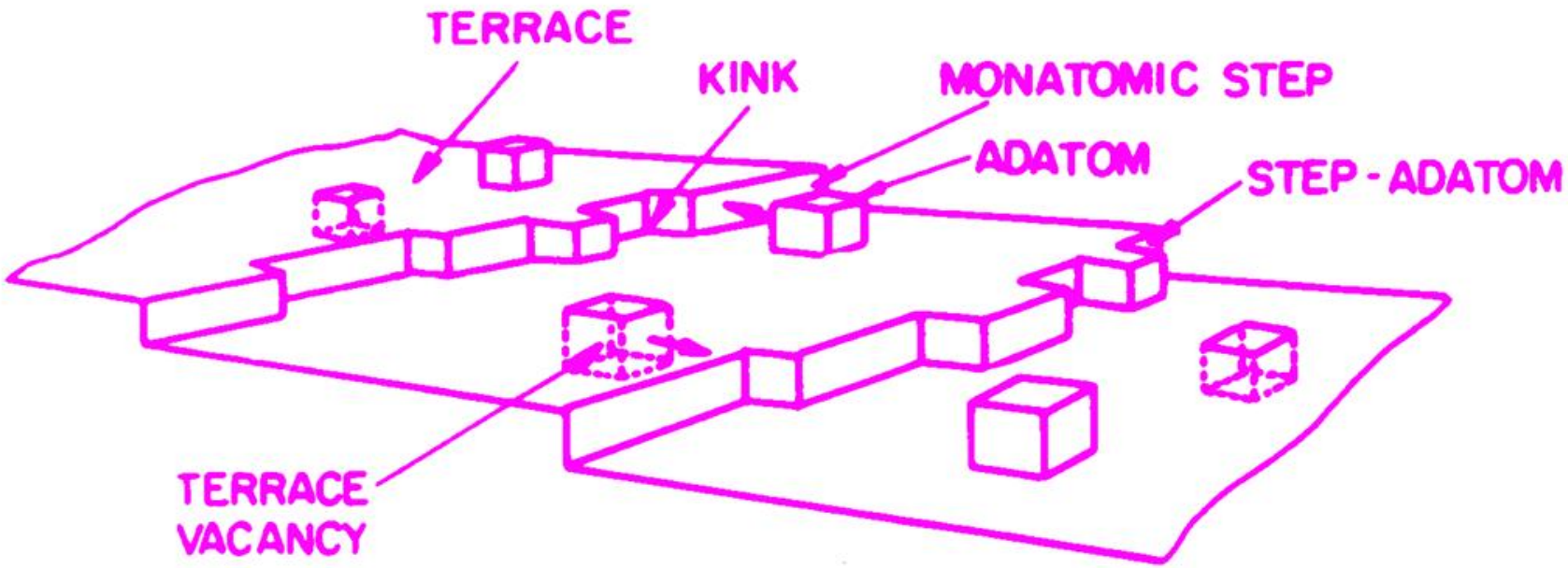


Figure 2.4. Scanning tunneling microscope (STM) picture of the (0001) face of rhenium over a 4000-$Å^2$ area.

STM picture of the (0001) face of Re over a 4000-A^2 area

描述表面不均匀性常用的模型



2. Clean Surface

固体表面原子浓度为$10^{15}/cm^2$数量级，为保持表面清洁, 必须保持同表面碰撞的分子数低于某水准

根据气体分子运动论, 在一定T一定P下, 单位时间单位面积上流过的气体量(通量)为

$$F = \frac{P}{(2\pi m k_B T)^{1/2}} = \frac{N_A P}{[2\pi M(\text{mol/g}) RT]^{1/2}}$$

$$= 2.63\times10^{20}\frac{P\,(\text{Pa})}{[M(\text{mol/g})\ T]^{1/2}} = 3.51\times10^{22}\frac{P\,(\text{torr})}{[M(\text{mol/g})\ T]^{1/2}}$$

假定M= 28, T= 300 K, P= 3×10^{-6} torr

F= 1.15×10^{15} molecule cm^{-2} s^{-1}

UHV要求

假定: M= 28, T= 300 K, P= 3 × 10^{-6} torr

$F \approx 10^{15}$ molecule cm^{-2} s^{-1}

换言之, 在P >3 × 10^{-6} torr 时, 如果每一次碰撞都可使分子在表面吸附的话, 固体表面在1s之内就被吸附物所覆盖。因而要在研究的时间内，如1h，保持表面清洁，则真空度必须小于10^{-9} torr。即研究清洁表面时，必须保持ultrahigh vacuum conditions.

研究气体分子在清洁表面吸附时，常使用表面覆盖度的概念，单位用Langmuir (L)

1 L= 10^{-6} torr s

1.4 Methods for Surface Characterizations

1 Adsorption
2 LEED
3 XPS
4 AES
5 EELS (HREELS)
6 EXAFS

Probe: molecule; atom; electron; X-ray; etc.

What information?

geometric structure;
electronic structure/chemical state;
composition; etc.

Why surface sensitive?

1. Adsorption

包括物理和化学吸附，其中物理吸附适用于所有多孔材料。

Probe: molecule

物理吸附的可获得信息为：

表面积、孔道特性

化学吸附的可获得信息为：

表面位浓度及分散度

要求：合适的真空度

2. Low-Energy Electron Diffraction (LEED)

从低能电子（≤500 ev电子束,平均自由程(波长) ≤ 2 nm ）与表面作用，发生弹性散射研究清洁固体表面或表面吸附的几何结构的有效手段。

Probe: electron

可获得信息为：

清洁表面原子排列形式（几何结构）

吸附原子或分子在表面排列形式

要求：超高真空



C.J. Davisson
G.P.Thomson
(1927年实验）
电子衍射实验贡献
1937年获Nobel物理学奖

3. X-ray Photoelectron Spectroscopy (XPS)

X射线激发固体中原子或离子的内层电子，通过能量差得出内层电子结合能的信息。对于特定的原子其结合能是特定的，因而可用于表面组成分析。随价态及化学环境的变化，结合能会有一定移动，从移动可以判断原子价态及配位环境。**因光电子逃逸深度小于2－3 nm, 所以是表面敏感手段。**

Probe: X-ray; photoelectron

可获得信息为：

表面组成

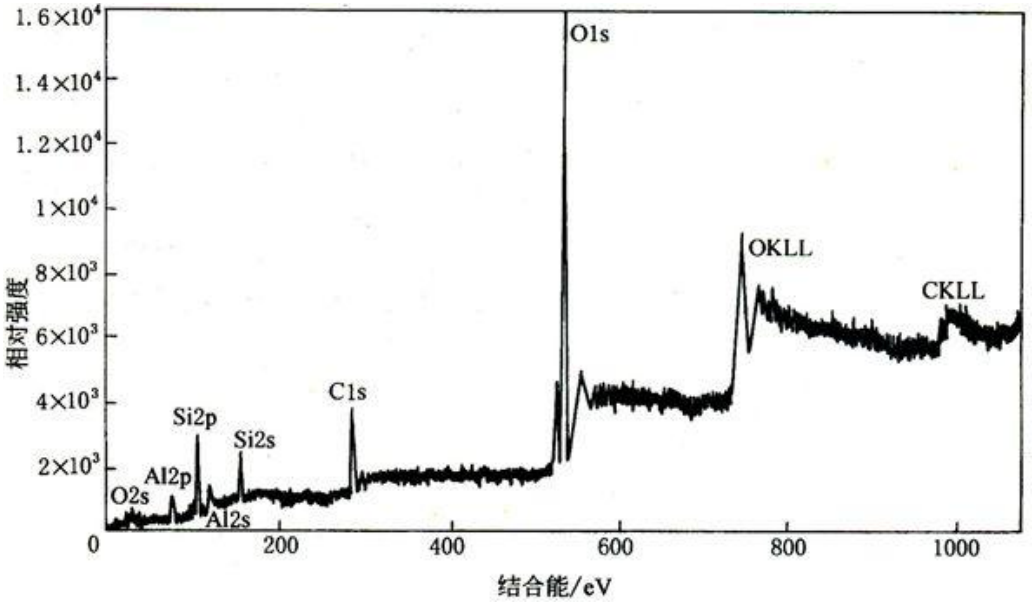
表面原子的价态、配位环境

要求：超高真空



K. M. Siegbahn 因对电子能谱学的贡献1981年获Nobel物理学奖





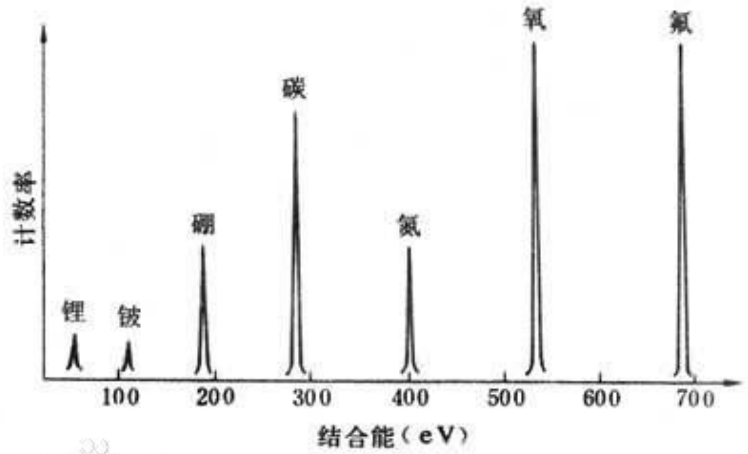


图 1　第二周期元素的 1s 电子结合能

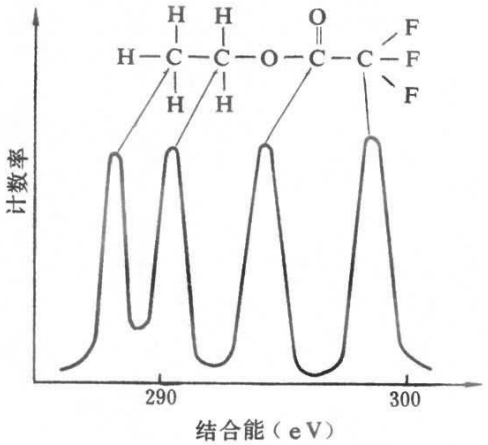


图 2　三氟乙酸乙酯中碳的 1s电子X射线光电子能谱

4. Auger Electron Spectroscopy (AES)

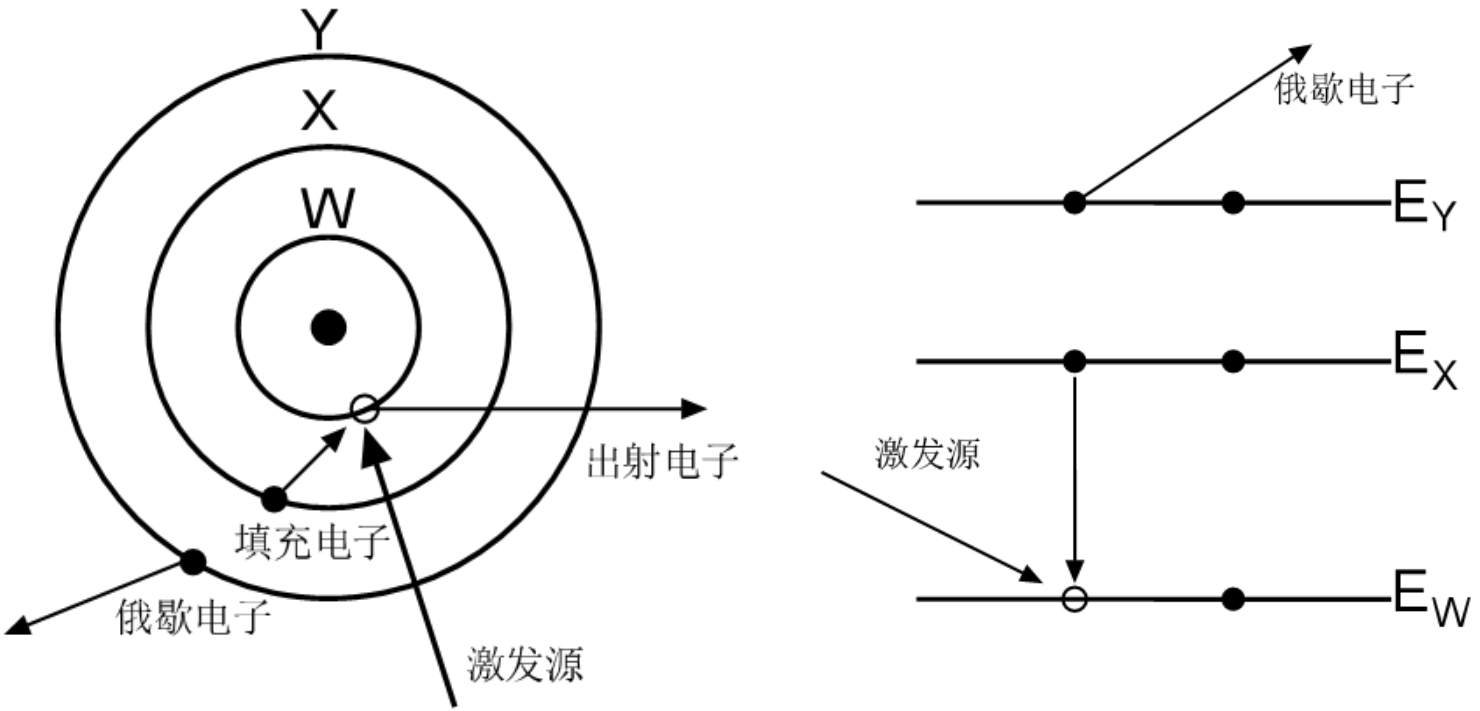
当一个内层的芯(K)电子被激发或电离时，其空穴会被一个从较高能级(L1)上下来的电子充填，产生的光子能量能进一步使外层的电子电离，产生的电子称Auger电子，该双电子过程称Auger过程。

Probe: X-ray; photoelectron

可获得信息为：

表面化学组成定量分析

要求：超高真空


Y
X
W
出射电子
填充电子
俄歇电子
激发源
俄歇电子
E_Y
E_X
激发源
E_W

5. High-Resolution Electron Energy Loss Spectroscopy (HREELS)

低能电子束同固体表面相互作用，引起表面吸附物种分子的振动发生跃迁，通过测定反射电子束的能量，从能量损失的情况得到吸附物种振动跃迁的信息。

HREELS实际上所得信息与IR相似，只不过HREELS使用低能电子，从而所得信息是**高度表面灵敏的。对表面吸附物种的灵敏程度为IR的约100倍。**

Probe: electron

可获得信息为：

表面吸附物种的键合结构

要求：超高真空

6. X-ray Absorption Fine Structure Spectroscopy (XAFS)

当X射线透过物质时，如果能激发内层电子就会被吸收。当X射线能量与电子结合能相同时，其吸收为最大，若X射线能量大于电子结合能时，对单个原子系统，吸收会单调减小。但在固体中由于受邻近原子的影响，吸收会随能量增加发生摆动，通过研究吸收系数的摆动规律，可以得到中心原子邻近的原子的配位数及键长度的信息。

Probe: X-ray

可获得信息为：

XAFS

EXAFS

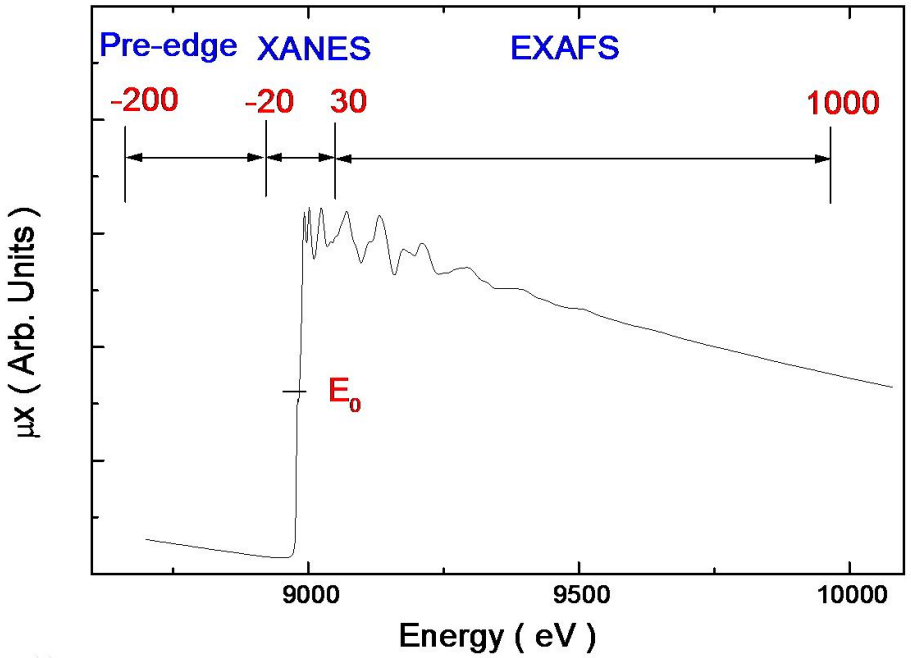
短程有序结构测定

纳米尺度材料结构表征

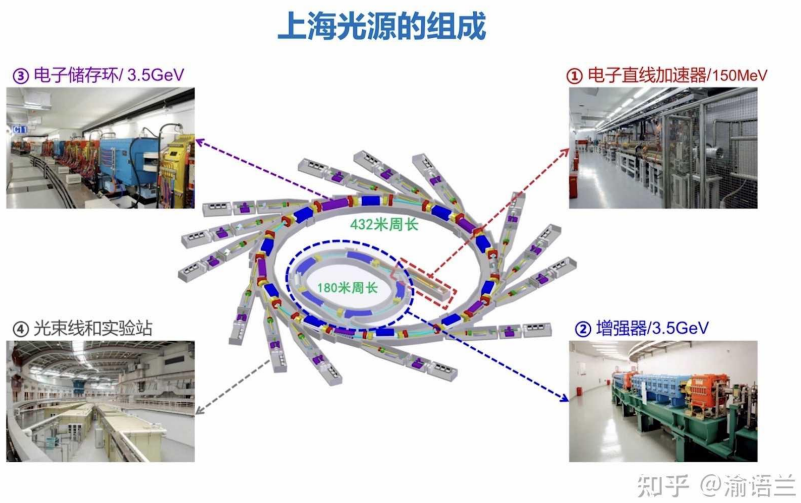
中心原子配位数及键距离

XANES (X-ray absorption near-edge structure spectroscopy)

无高真空的要求，适合于原位研究


Pre-edge
XANES
EXAFS
-200
-20
30
1000
μx (Arb. Units)
E0
9000
9500
10000
Energy (eV)




上海光源的组成
③ 电子储存环/ 3.5GeV
① 电子直线加速器/150MeV
432米周长
180米周长
④ 光束线和实验站
② 增强器/3.5GeV
知乎 @渝语兰

上海同步辐射光源

Surface Science Techniques for Molecular-Level Studies

Surface Composition:

Auger electron spectroscopy (AES), X-ray photoelectron spectroscopy (XPS) , Secondary ion mass spectrometry (SIMS), Ion scattering spectroscopy (ISS)

Surface Structure:

Low-energy electron diffraction (LEED), Transmission electron microscopy (TEM), Atom diffraction, Atomic force microscopy (AFM), Scanning Tunneling microscopy (STM), Surface X-ray diffraction, X-ray absorption fine structure (XAFS)

Surface Electron Structure:

Second harmonic generation (SHG), Ultraviolet Photoelectron Spectroscopy, X-ray emission spectroscopy, Scanning tunneling spectroscopy (STS)

Surface Vibration and Reaction Dynamics:

Electron energy loss spectroscopy (EELS), Helium atom scattering, IR spectroscopy, Raman spectroscopy, Sum-frequency generation spectroscopy (SFG), Molecular beam surface scattering